94. Fifth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ +\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 0.)}$$

1	0	0	1	$= E+G\cdot E'+G'$		$+$	$F+H\cdot F'+H'$	
5	4	4	5	$i.E-G$	„	$+$	$i.F-H$	„
9	8	8	9	$E+G$	„	$-$	$.F+H$	„
13	12	12	13	$i.E-G$	„	$-$	$i.F-H$	„
3	2	2	3	$F+H$	„	$+$	$E+G$	„
7	6	6	7	$i.F-H$	„	$+$	$i.E-G$	„
11	10	10	11	$-.F+H$	„	$+$	$E+G$	„
15	14	14	15	$-i.F-H$	„	$+$	$i.E-G$	„

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ -\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 0.)}$$

1	0	0	1	$= E-G\cdot E'-G'$		$+$	$F-H\cdot F'-H'$	
5	4	4	5	$i.E+G$	„	$+$	$i.F+H$	„
9	8	8	9	$E-G$	„	$-$	$.F-H$	„
13	12	12	13	$i.E+G$	„	$-$	$i.F+H$	„
3	2	2	3	$F-H$	„	$+$	$E-G$	„
7	6	6	7	$i.F+H$	„	$+$	$i.E+G$	„
11	10	10	11	$-.F-H$	„	$+$	$E-G$	„
15	14	14	15	$-i.F+H$	„	$+$	$i.E+G$	„

95. Sixth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ +\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 1.)}$$

5	0	0	5	$= E-iG\cdot E'+iG'$		$+$	$F-iH\cdot F'+iH'$	
1	4	4	1	$-i.E+iG$	„	$-$	$i.F+iH$	„
13	8	8	13	$E-iG$	„	$-$	$.F-iH$	„
9	12	12	9	$-i.E+iG$	„	$+$	$i.F+iH$	„
7	2	2	7	$F-iH$	„	$+$	$E-iG$	„
3	6	6	3	$-i.F+iH$	„	$-$	$i.E+iG$	„
15	10	10	15	$-.F-iH$	„	$+$	$E-iG$	„
11	14	14	11	$i.F+iH$	„	$-$	$i.E+iG$	„

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ -\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 1.)}$$

5	0	0	5	$= E+iG\cdot E'-iG'$		$+$	$F+iH\cdot F'-iH'$	
1	4	4	1	$-i.E-iG$	„	$-$	$i.F-iH$	„
13	8	8	13	$E+iG$	„	$-$	$.F+iH$	„
9	12	12	9	$-i.E-iG$	„	$+$	$i.F-iH$	„
7	2	2	7	$F+iH$	„	$+$	$E+iG$	„
3	6	6	3	$-i.F-iH$	„	$-$	$i.E-iG$	„
15	10	10	15	$-.F+iH$	„	$+$	$E+iG$	„
11	14	14	11	$i.F-iH$	„	$-$	$i.E-iG$	„

96. Seventh set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ +\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad \text{(Suffixes 2.)}$$

9	0	0	9	$= E+G\cdot E'+G'$	$+$	$F-H\cdot F'-H'$
13	4	4	13	$i.E-G$ „	$+$	$i.F+H$ „
1	8	8	1	$E+G$ „	$-$	$.F-H$ „
5	12	12	5	$i.E-G$ „	$-$	$i.F+H$ „
11	2	2	11	$F+H$ „	$+$	$E-G$ „
15	6	6	15	$i.F-H$ „	$+$	$i.E+G$ „
3	10	10	3	$F+H$ „	$-$	$.E-G$ „
7	14	14	7	$i.F-H$ „	$-$	$i.E+G$ „

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ -\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad \text{(Suffixes 2.)}$$

9	0	0	9	$= E-G\cdot E'-G'$	$+$	$F+H\cdot F'+H'$
13	4	4	13	$i.E+G$ „	$+$	$i.F-H$ „
1	8	8	1	$E-G$ „	$-$	$.F+H$ „
5	12	12	5	$i.E+G$ „	$-$	$i.F-H$ „
11	2	2	11	$F-H$ „	$+$	$E+G$ „
15	6	6	15	$i.F+H$ „	$+$	$i.E-G$ „
3	10	10	3	$F-H$ „	$-$	$.E+G$ „
7	14	14	7	$i.F+H$ „	$-$	$i.E-G$ „

97. Eighth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ +\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad \text{(Suffixes 3.)}$$

13	0	0	13	$= E-iG\cdot E'+iG'$	$+$	$F+iH\cdot F'-iH'$
9	4	4	9	$-i.E+iG$ „	$-$	$i.F-iH$ „
5	8	8	5	$E-iG$ „	$-$	$.F+iH$ „
1	12	12	1	$-i.E+iG$ „	$+$	$i.F-iH$ „
15	2	2	15	$F-iH$ „	$+$	$E+iG$ „
11	6	6	11	$-i.F+iH$ „	$-$	$i.E-iG$ „
7	10	10	7	$F-iH$ „	$-$	$.E+iG$ „
3	14	14	3	$-i.F+iH$ „	$+$	$i.E-iG$ „

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ -\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad \text{(Suffixes 3.)}$$

13	0	0	13	$= E+iG\cdot E'-iG'$	$+$	$F-iH\cdot F'+iH'$
9	4	4	9	$-i.E-iG$ „	$-$	$i.F+iH$ „
5	8	8	5	$E+iG$ „	$-$	$.F-iH$ „
1	12	12	1	$-i.E-iG$ „	$+$	$i.F+iH$ „
15	2	2	15	$F+iH$ „	$+$	$E-iG$ „
11	6	6	11	$-i.F-iH$ „	$-$	$i.E+iG$ „
7	10	10	7	$F+iH$ „	$-$	$.E-iG$ „
3	14	14	3	$-i.F-iH$ „	$+$	$i.E+iG$ „

98. Ninth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3} + \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\} \qquad \text{(Suffixes 0.)}$$

2	0	0	2	$= I+K\cdot I'+K'$	$+$	$J+L\cdot J'+L'$
6	4	4	6	$I+K$ „	$-$	$.J+L$ „
10	8	8	10	$i.I-K$ „	$+$	$i.J-L$ „
14	12	12	14	$i.I-K$ „	$-$	$i.J-L$ „
3	1	1	3	$J+L$ „	$+$	$I+K$ „
7	5	5	7	$-.J+L$ „	$-$	$.I+K$ „
11	9	9	11	$i.J-L$ „	$+$	$i.I-K$ „
15	13	13	15	$-i.J-L$ „	$-$	$i.I-K$ „

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3} - \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\} \qquad \text{(Suffixes 0.)}$$

2	0	0	2	$= I-K\cdot I'-K'$	$+$	$J-L\cdot J'-L'$
6	4	4	6	$I-K$ „	$-$	$.J-L$ „
10	8	8	10	$i.I+K$ „	$+$	$i.J+L$ „
14	12	12	14	$i.I+K$ „	$-$	$i.J+L$ „
3	1	1	3	$J-L$ „	$+$	$I-K$ „
7	5	5	7	$-.J-L$ „	$+$	$I-K$ „
11	9	9	11	$i.J+L$ „	$+$	$i.I+K$ „
15	13	13	15	$-i.J+L$ „	$+$	$i.I+K$ „

99. Tenth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3} + \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\} \qquad \text{(Suffixes 1.)}$$

6	0	0	6	$= I+K\cdot I'+K'$	$+$	$J-L\cdot J'-L'$
2	4	4	2	$I+K$ „	$-$	$.J-L$ „
14	8	8	14	$i.I-K$ „	$+$	$i.J+L$ „
10	12	12	10	$i.I-K$ „	$-$	$i.J+L$ „
7	1	1	7	$J+L$ „	$+$	$I-K$ „
3	5	5	3	$J+L$ „	$-$	$.I-K$ „
15	9	9	15	$i.J-L$ „	$+$	$i.I+K$ „
11	13	13	11	$i.J-L$ „	$-$	$i.I+K$ „

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3} - \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\} \qquad \text{(Suffixes 1.)}$$

6	0	0	6	$= I-K\cdot I'-K'$	$+$	$J+L\cdot J'+L'$
2	4	4	2	$I-K$ „	$-$	$.J+L$ „
14	8	8	14	$i.I+K$ „	$+$	$i.J-L$ „
10	12	12	10	$i.I+K$ „	$-$	$i.J-L$ „
7	1	1	7	$J-L$ „	$+$	$I+K$ „
3	5	5	3	$J-L$ „	$-$	$.I+K$ „
15	9	9	15	$i.J+L$ „	$+$	$i.I-K$ „
11	13	13	11	$i.J+L$ „	$-$	$i.I-K$ „

100. Eleventh set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ +\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad \text{(Suffixes 2.)}$$

10	0	0	10	$= I - iK \cdot I' + iK'$	$+$	$J - iL \cdot J' + iL'$
14	4	4	14	$I - iK$,,	$-$	$.J - iL$,,
2	8	8	2	$-i.I + iK$,,	$-$	$i.J + iL$,,
6	12	12	6	$-i.I + iK$,,	$+$	$i.J + iL$,,
11	1	1	11	$J - iL$,,	$+$	$I - iK$,,
15	5	5	15	$-.J - iL$,,	$+$	$I - iK$,,
3	9	9	3	$-i.J + iL$,,	$-$	$i.I + iK$,,
7	13	13	7	$i.J + iL$,,	$-$	$i.I + iK$,,

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ -\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad \text{(Suffixes 2.)}$$

10	0	0	10	$= I + iK \cdot I' - iK'$	$+$	$J + iL \cdot J' - iL'$
14	4	4	14	$I + iK$,,	$-$	$.J + iL$,,
2	8	8	2	$-i.I - iK$,,	$-$	$i.J - iL$,,
6	12	12	6	$-i.I - iK$,,	$+$	$i.J - iL$,,
11	1	1	11	$J + iL$,,	$+$	$I + iK$,,
15	5	5	15	$-.J + iL$,,	$+$	$I + iK$,,
3	9	9	3	$-i.J - iL$,,	$-$	$i.I - iK$,,
7	13	13	7	$i.J - iL$,,	$-$	$i.I - iK$,,

101. Twelfth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ +\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad \text{(Suffixes 3.)}$$

14	0	0	14	$= I - iK \cdot I' + iK'$	$+$	$J + iL \cdot J' - iL'$
10	4	4	10	$I - iK$,,	$-$	$.J + iL$,,
6	8	8	6	$-i.I + iK$,,	$-$	$i.J - iL$,,
2	12	12	2	$-i.I + iK$,,	$+$	$i.J - iL$,,
15	1	1	15	$J - iL$,,	$+$	$I + iK$,,
11	5	5	11	$J - iL$,,	$-$	$.I + iK$,,
7	9	9	7	$-i.J + iL$,,	$-$	$i.I - iK$,,
3	13	13	3	$-i.J + iL$,,	$+$	$i.I - iK$,,

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ -\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad \text{(Suffixes 3.)}$$

14	0	0	14	$= I + iK \cdot I' - iK'$	$+$	$J - iL \cdot J' + iL'$
10	4	4	10	$I + iK$,,	$-$	$.J - iL$,,
6	8	8	6	$-i.I - iK$,,	$-$	$i.J + iL$,,
2	12	12	2	$-i.I - iK$,,	$+$	$i.J + iL$,,
15	1	1	15	$J + iL$,,	$+$	$I - iK$,,
11	5	5	11	$J + iL$,,	$-$	$.I - iK$,,
7	9	9	7	$-i.J - iL$,,	$-$	$i.I + iK$,,
3	13	13	3	$-i.J - iL$,,	$+$	$i.I + iK$,,

102. Thirteenth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ +\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 0.)}$$

3	0	0	3	$= M+Q\cdot M'+Q'$		$+$	$N+P\cdot N'+P'$	
7	4	4	7	$i.M-Q$	„	$-$	$i.N-P$	„
11	8	8	11	$i.M-Q$	„	$+$	$i.N-P$	„
15	12	12	15	$-.M+Q$	„	$+$	$N+P$	„
2	1	1	2	$N+P$	„	$+$	$M+Q$	„
6	5	5	6	$-i.N-P$	„	$+$	$i.M-Q$	„
10	9	9	10	$i.N-P$	„	$+$	$i.M-Q$	„
14	13	13	14	$N+P$	„	$-$	$.M+Q$	„

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ -\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 0.)}$$

3	0	0	3	$= M-Q\cdot M'-Q'$		$+$	$N-P\cdot N'-P'$	
7	4	4	7	$i.M+Q$	„	$-$	$i.N+P$	„
11	8	8	11	$i.M+Q$	„	$+$	$i.N+P$	„
15	12	12	15	$-.M-Q$	„	$+$	$N-P$	„
2	1	1	2	$N-P$	„	$+$	$M-Q$	„
6	5	5	6	$-i.N+P$	„	$+$	$i.M+Q$	„
10	9	9	10	$i.N+P$	„	$+$	$i.M+Q$	„
14	13	13	14	$N-P$	„	$-$	$.M-Q$	„

103. Fourteenth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ +\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 1.)}$$

7	0	0	7	$= M-iQ\cdot M'+iQ'$		$+$	$N+iP\cdot N'-iP'$	
3	4	4	3	$-i.M+iQ$	„	$+$	$i.N-iP$	„
15	8	8	15	$i.M+iQ$	„	$+$	$i.N-iP$	„
11	12	12	11	$M-iQ$	„	$-$	$.N+iP$	„
6	1	1	6	$N-iP$	„	$+$	$M+iQ$	„
2	5	5	2	$-i.N-iP$	„	$+$	$i.M-iQ$	„
14	9	9	14	$i.N+iP$	„	$+$	$i.M-iQ$	„
10	13	13	10	$N-iP$	„	$-$	$.M+iQ$	„

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\ -\ \vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 1.)}$$

7	0	0	7	$= M+iQ\cdot M'-iQ'$		$+$	$N-iP\cdot N'+iP'$	
3	4	4	3	$-i.M-iQ$	„	$+$	$i.N+iP$	„
15	8	8	15	$-i.M-iQ$	„	$+$	$i.N+iP$	„
11	12	12	11	$M+iQ$	„	$-$	$.N-iP$	„
6	1	1	6	$N+iP$	„	$+$	$M-iQ$	„
2	5	5	2	$-i.N+iP$	„	$+$	$i.M+iQ$	„
14	9	9	14	$i.N-iP$	„	$+$	$i.M+iQ$	„
10	13	13	10	$N+iP$	„	$-$	$.M-iQ$	„

104. Fifteenth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\;+\;\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 2.)}$$

11	0	0	11	$= M - iQ \cdot M' + iQ'$		$+$	$N - iP \cdot N' + iP'$	
15	4	4	15	$i.M + iQ$	,,	$-$	$i.N + iP$	,,
3	8	8	3	$-i.M + iQ$	,,	$-$	$i.N + iP$	,,
7	12	12	7	$M - iQ$	,,	$-$	$.N - iP$	,,
10	1	1	10	$N - iP$	,,	$+$	$M - iQ$	,,
14	5	5	14	$-i.N + iP$	,,	$+$	$i.M + iQ$	,,
2	9	9	2	$-i.N + iP$	,,	$-$	$i.M + iQ$	,,
6	13	13	6	$-.N - iP$	,,	$+$	$M - iQ$	,,

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\;-\;\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 2.)}$$

11	0	0	11	$= M + iQ \cdot M' - iQ'$		$+$	$N + iP \cdot N' - iP'$	
15	4	4	15	$i.M - iQ$	,,	$-$	$i.N - iP$	,,
3	8	8	3	$-i.M - iQ$	,,	$-$	$i.N - iP$	,,
7	12	12	7	$M + iQ$	,,	$-$	$.N + iP$	,,
10	1	1	10	$N + iP$	,,	$+$	$M + iQ$	,,
14	5	5	14	$-i.N - iP$	,,	$+$	$i.M - iQ$	,,
2	9	9	2	$-i.N - iP$	,,	$-$	$i.M - iQ$	,,
6	13	13	6	$-.N + iP$	,,	$+$	$M + iQ$	,,

105. Sixteenth set of 16.

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\;+\;\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 3.)}$$

15	0	0	15	$= M - Q \cdot M' - Q'$		$+$	$N + P \cdot N' + P'$	
11	4	4	11	$-i.M + Q$	,,	$+$	$i.N - P$	,,
7	8	8	7	$-i.M + Q$	,,	$-$	$i.N - P$	,,
3	12	12	3	$-.M - Q$	,,	$+$	$N + P$	,,
14	1	1	14	$N - P$	,,	$+$	$M + Q$	,,
10	5	5	10	$-i.N + P$	,,	$+$	$i.M - Q$	,,
6	9	9	6	$-i.N + P$	,,	$-$	$i.M - Q$	,,
2	13	13	2	$-.N - P$	,,	$+$	$M + Q$	,,

$$\tfrac{1}{2}\left\{\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\;-\;\vartheta\frac{u+u'}{3}\cdot\vartheta\frac{u-u'}{3}\right\}\qquad\text{(Suffixes 3.)}$$

15	0	0	15	$= M + Q \cdot M' + Q'$		$+$	$N - P \cdot N' - P'$	
11	4	4	11	$-i.M - Q$	,,	$+$	$i.N + P$	,,
7	8	8	7	$-i.M - Q$	,,	$-$	$i.N + P$	,,
3	12	12	3	$-.M + Q$	,,	$+$	$N - P$	,,
14	1	1	14	$N + P$	,,	$+$	$M - Q$	,,
10	5	5	10	$-i.N - P$	,,	$+$	$i.M + Q$	,,
6	9	9	6	$-i.N - P$	,,	$-$	$i.M + Q$	,,
2	13	13	2	$-.N + P$	,,	$+$	$M - Q$	,,

106. In the square set, writing $u' = v' = 0$, and $\alpha, \beta, \gamma, \delta$ for X', Y', Z', W', also slightly altering the arrangement, the system becomes: and further writing herein $u = 0,\ v = 0$, it becomes

$\vartheta^2 u$	X	Y	Z	W	$\vartheta^2 0$	
4	α	β	γ	δ	$4 = \alpha^2 + \beta^2 + \gamma^2 + \delta^2$	$= 4$
8	α	$-\beta$	γ	$-\delta$	$8 = \alpha^2 - \beta^2 + \gamma^2 - \delta^2$	$= 8$
12	α	β	$-\gamma$	$-\delta$	$12 = \alpha^2 - \beta^2 - \gamma^2 + \delta^2$	$= 12$
1	β	α	δ	γ	$1 = 2(\alpha\beta + \gamma\delta)$	$= 1$
5	β	$-\alpha$	δ	$-\gamma$	5	
9	β	α	$-\delta$	$-\gamma$	$9 = 2(\alpha\beta - \gamma\delta)$	$= 9$
13	β	$-\alpha$	$-\delta$	γ	13	
2	γ	δ	α	β	$2 = 2(\alpha\gamma + \beta\delta)$	$= 2$
6	γ	$-\delta$	α	$-\beta$	$6 = 2(\alpha\gamma - \beta\delta)$	$= 6$
10	γ	$-\alpha$	$-\beta$	δ	10	
14	γ	$-\delta$	$-\alpha$	β	14	
3	δ	γ	β	α	$3 = 2(\alpha\delta + \beta\gamma)$	$= 3$
7	δ	$-\gamma$	β	$-\alpha$	7	
11	δ	γ	$-\beta$	$-\alpha$	11	
15	δ	$-\gamma$	$-\beta$	α	$15 = 2(\alpha\delta - \beta\gamma)$	$= 15$

viz. this last is the before-mentioned system of equations giving the values of the 10 zero-functions c in terms of the four constants $\alpha, \beta, \gamma, \delta$.

107. The system first obtained is a system of 16 equations

$$\vartheta^2(u, v) = \alpha X + \beta Y + \gamma Z + \delta W, \text{ \&c.,}$$

showing that the squares of the theta-functions are each of them a linear function of the four quantities X, Y, Z, W. If the functions on the right-hand side were independent (asyzygetic) linear functions of X, Y, Z, W, it would follow that any four (selected at pleasure) of the squared theta-functions are linearly independent, and that we could in terms of these four express linearly each of the remaining 12 squared functions. But this is not so; the form of the linear functions of (X, Y, Z, W) is such that we can (and that in 16 different ways) select out of the 16 linear functions six functions, such that any four of them are connected by a linear equation; and there are consequently 16 hexads of squared theta-functions, such that any four out of the same hexad are connected by a linear relation. The hexads are shown by the foregoing "Table of the 16 Kummer hexads."

108. The *à posteriori* verification is immediately effected; taking for instance the first column, the equations are

	X	Y	Z	W
A, 11	β	γ	$-\beta$	$-\alpha$
B, 7	δ	$-\gamma$	β	$-\alpha$
AB, 6	γ	$-\delta$	α	$-\beta$
CD, 2	γ	δ	α	β
CE, 1	β	α	δ	γ
DE, 9	β	α	$-\delta$	$-\gamma$

viz. it should thence follow that there is a linear relation between any four of the six squared functions 11, 7, 6, 2, 1, 9: and it is accordingly seen that this is so. It further appears that, in the several linear relations, the coefficients (obtained in the first instance as functions of $\alpha, \beta, \gamma, \delta$) are in fact the 10 constants c: the 15 relations connecting the several systems of four out of the six squared functions are given in the following table.

109.

ϑ^2	11	7	6	2	1	9	$=0$
c_0^2			6	-2	1	-9	
c_2^2		6		$+15$	-12	$+4$	
c_4^2	-2		-15		$+8$	-3	
c_5^2		1	$+12$	-8		$+3$	
c_6^2	-9		-4	$+3$			
c_8^2		6		3		$+8$	
c_9^2		-2	-3		$+4$	-12	
c_{10}^2		1	$+8$	-4		-15	
c_{11}^2		-9	-8	$+12$	$+15$		
c_{12}^2	-15	$+3$	$+2$	-6			
c_{13}^2	-12		$+1$		-6		
c_{14}^2	-4	$+8$	$+9$			-6	
c_{15}^2	-3	$+15$			$+9$	-1	
c_3^2	-2	$+12$		-9			
c_7^2	-8	$+4$		$+1$	-2		

Read

$$c_2^4\,\vartheta_{11}^2 + c_4^2\,\vartheta_6^2 - c_5^2\,\vartheta_2^2 + c_6^2\,\vartheta_1^2 - c_8^2\,\vartheta_9^2 = 0, \;\; \&c.$$

110. The first set of 16 equations is the square-set, which has been already considered. If in each of the other sets of 16 equations we write in like manner $u' = 0$, each set in fact reduces itself to eight equations; sets 2, 3, 4 give thus $8+8+8 = 24$ equations; sets 5 to 8, 9 to 12, and 13 to 15, give each $8+8+8+8 = 32$ equations; or we have sets of 24, 32, 32, 32, together 120 equations, the number being of course one half of $256 - 16$, the number of equations after deducting the 16 equations of the square set.

111. The first set, 24 equations.

This is derived from the second, third, and fourth sets, each of 16 equations, by writing therein $u' = 0$. Taking $\alpha_1, \beta_1, \gamma_1, \delta_1$ for the zero-functions corresponding to X_1, Y_1, Z_1, W_1, then on writing $u' = 0$, X_1', Y_1', Z_1', W_1' become $\alpha_1, \beta_1, \gamma_1, \delta_1$. In the second set of 16 equations, the first equations thus are

$$\vartheta_1 u \cdot \vartheta_1 u = \alpha_1 X_1 + \gamma_1 Z_1 = \beta_1 Y_1 + \delta_1 W_1,$$

$$\vartheta_9 u \cdot \vartheta_8 u = \alpha_1 X_1 - \gamma_1 Z_1, \quad 0 = \beta_1 Y_1 - \delta_1 W_1,$$

viz. the equations of the column require that, and are all satisfied if, $\beta_1 = 0$, $\delta_1 = 0$; hence the zero-functions are $\alpha_1, 0, \gamma_1, 0$; and this being so we have only the equations of the first column. And similarly as regards the third and fourth sets the zero-values corresponding to

$$X_1, Y_1, Z_1, W_1 \quad | \quad X_2, Y_2, Z_2, W_2 \quad | \quad X_3, Y_3, Z_3, W_3$$

are

$$\alpha_1,\ 0,\ \gamma_1,\ 0 \ \ | \ \ \alpha_2,\ 0,\ 0,\ \delta_2 \ \ | \ \ \alpha_3,\ 0,\ 0,\ \delta_3;$$

and we have in all $8+8+8 = 24$ equations. These are

(Suffixes 1.)					(Suffixes 2.)					(Suffixes 3.)				
			X	Z				X	W				X	W
4		=	α	γ	8		=	α	δ	12		=	α	δ
12	8	=	α	$-\gamma$	12	4	=	α	$-\delta$	8	4	=	α	$-\delta$
6	2	=	γ	α	9		=	δ	α	15	3	=	δ	α
14	10	=	γ	$-\alpha$	13		=	$-\delta$	α	11	7	=	$-\delta$	α
			Y	W				Y	W				Y	Z
5	1	=	α	γ	10	2	=	α	β	13	1	=	α	δ
13	9	=	α	$-\gamma$	14	6	=	α	$-\beta$	9	5	=	α	$-\delta$
7	3	=	γ	α	11	3	=	β	α	14	2	=	δ	α
15	11	=	γ	$-\alpha$	15	7	=	β	$-\alpha$	10	6	=	$-\delta$	α
			$\vartheta^2 0$					$\vartheta^2 0$					$\vartheta^2 0$	
4		=	$\alpha^2+\gamma^2$		8		=	$\alpha^2+\delta^2$		12		=	$\alpha^2+\delta^2$	
12	8	=	$\alpha^2-\gamma^2$		12	4	=	$\alpha^2-\delta^2$		8	4	=	$\alpha^2-\delta^2$	
6	2	=	$2\alpha\gamma$		9		=	$2\alpha\beta$		15	3	=	$2\alpha\delta$.	

112. The second set, 32 equations.

To exhibit these in a convenient form, I alter the notation, viz. I write

	$E+G$	$i(E-G)$	$(F+H)$	$i(F-H)$	E_1+iG_1	E_1-iG_1	F_1+iH_1	F_1-iH_1
$=$	X	Y	Z	W	X_1	Y_1	Z_1	W_1
	(E_2+G_2)	$i(E_2-G_2)$	(F_2+H_2)	$i(F_2-H_2)$	E_3+iG_3	E_3-iG_3	F_3+iH_3	F_3-iH_3
$=$	X_2	Y_2	Z_2	W_2	X_3	Y_3	Z_3	W_3

so that as regards the present set of equations, X, Y, Z, W, signify as just mentioned. And, this being so, the corresponding zero-values are

$$\alpha,\ 0,\ \gamma,\ 0\ \mid\ \alpha_1,\ 0,\ \gamma_1,\ 0\ \mid\ \alpha_2,\ 0,\ 0,\ \delta_2\ \mid\ \alpha_3,\ 0,\ 0,\ \delta_3.$$

The equations then are

(Suffixes 0.)					(Suffixes 1.)					(Suffixes 2.)					(Suffixes 3.)				
			X	Z				X	Z				X	W				X	W
1		$=$	α	γ	1	4	$=$	$-i\alpha$	$-i\gamma$	9		$=$	α	δ	9	4	$=$	$-i\alpha$	$-i\delta$
9	8	$=$	α	$-\gamma$	9	12	$=$	$-i\alpha$	$+i\gamma$	1	8	$=$	α	$-\delta$	1	12	$=$	$-i\alpha$	$+i\delta$
3	2	$=$	γ	α	3	6	$=$	$-i\gamma$	$-i\alpha$	15	6	$=$	δ	α	15	2	$=$	δ	α
11	10	$=$	γ	$-\alpha$	11	14	$=$	$-i\gamma$	$+i\alpha$	7	14	$=$	$-\delta$	α	7	10	$=$	$-\delta$	α
			Y	W				Y	W				Y	Z				Y	Z
5	4	$=$	α	γ	5		$=$	α	γ	13	4	$=$	α	δ	13		$=$	α	δ
13	12	$=$	α	$-\gamma$	13	8	$=$	α	$-\gamma$	5	12	$=$	α	$-\delta$	5	8	$=$	α	$-\delta$
7	6	$=$	γ	α	7	2	$=$	γ	α	11	2	$=$	$-\delta$	α	11	6	$=$	$-i\delta$	$-i\alpha$
15	14	$=$	γ	$-\alpha$	15	10	$=$	γ	$-\alpha$	3	10	$=$	δ	α	3	14	$=$	$i\delta$	$-i\alpha$
			$\vartheta^2 0$					$\vartheta^2 0$					$\vartheta^2 0$					$\vartheta^2 0$	
1		$=$	$\alpha^2+\gamma^2$		1	4	$=$	$-i(\alpha^2+\gamma^2)$		9		$=$	$\alpha^2-\delta^2$		9	4	$=$	$-i(\alpha^2+\delta^2)$	
9	8	$=$	$\alpha^2-\gamma^2$		9	12	$=$	$-i(\alpha^2-\gamma^2)$		1	8	$=$	$\alpha^2+\delta^2$		1	12	$=$	$-i(\alpha^2-\delta^2)$	
3	2	$=$	$2\alpha\gamma$		3	6	$=$	$-2i\alpha\gamma$		15	6	$=$	$2\alpha\delta$		15	2	$=$	$2\alpha\delta.$	

113. Third set, 32 equations.

We again change the notation, writing

$$\begin{array}{c|cccc|cccc}
 & I+K & i(I-K) & J+L & i(J-L) & I_1+iK_1 & I_1-iK_1 & J_1+iL_1 & J_1-iL_1 \\
= & X & Y & Z & W & X_1 & Y_1 & Z_1 & W_1 \\
\hline
 & I_2+iK_2 & I_2-iK_2 & J_2+iL_2 & J_2-iL_2 & I_3+iK_3 & I_3-iK_3 & J_3+iL_3 & J_3-iL_3 \\
= & X_2 & Y_2 & Z_2 & W_2 & X_3 & Y_3 & Z_3 & W_3
\end{array}$$

the zero values being

$$\alpha,\ 0,\ \gamma,\ 0\ \mid\ \alpha_1,\ 0,\ 0,\ \delta_1\ \mid\ \alpha_2,\ 0,\ \gamma_2,\ 0\ \mid\ \alpha_3,\ 0,\ 0,\ \delta_3.$$

Then equations are

(Suffixes 0.)				
			X	Z
2		=	α	γ
6	4	=	α	$-\gamma$
3	1	=	γ	α
7	5	=	γ	$-\alpha$
			Y	W
10	8	=	α	γ
14	12	=	α	$-\gamma$
11	9	=	γ	α
15	13	=	γ	$-\alpha$
			$\vartheta^2 0$	
2		=	$\alpha^2+\gamma^2$	
6	4	=	$\alpha^2-\gamma^2$	
3	1	=	$2\alpha\gamma$	

(Suffixes 1.)				
			X	W
6		=	α	δ
2	4	=	α	$-\delta$
15	9	=	δ	α
11	13	=	$-\delta$	α
			Y	Z
			$\vartheta^2 0$	
6		=	$\alpha^2+\delta^2$	
2	4	=	$\alpha^2+\delta^2$	
15	9	=	$2\alpha\delta$	

(Suffixes 2.)				
			X	Z
2	8	=	$-i\alpha$	$-i\gamma$
6	12	=	$-i\alpha$	$+i\gamma$
3	9	=	$-i\gamma$	$-i\alpha$
7	13	=	$-i\gamma$	$+i\alpha$
			Y	W
10		=	α	γ
14	4	=	α	$-\gamma$
11	1	=	γ	α
15	5	=	γ	$-\alpha$
			$\vartheta^2 0$	
2	8	=	$-i(\alpha^2+\gamma^2)$	
6	12	=	$-i(\alpha^2-\gamma^2)$	
3	9	=	$-2i\alpha\gamma$	

(Suffixes 3.)				
			X	W
6	8	=	$-i\alpha$	$-i\delta$
2	12	=	$-i\alpha$	$+i\delta$
15	1	=	δ	α
11	5	=	$-\delta$	α
			Y	Z
14		=	α	δ
10	4	=	α	$-\delta$
7	9	=	$-i\delta$	$-i\alpha$
3	13	=	$i\delta$	$-i\alpha$
			$\vartheta^2 0$	
6	8	=	$-i(\alpha^2+\delta^2)$	
2	12	=	$-i(\alpha^2-\delta^2)$	
15	1	=	$2\alpha\delta.$	

114. Fourth set, 32 equations.

Again changing the notation, we write

$$\begin{array}{c|cccc|cccc} & M+Q & i(M-Q) & N+P & i(N-P) & M_1+iQ_1 & M_1-iQ_1 & N_1+iP_1 & N_1-iP_1 \\ = & X & Y & Z & W & X_1 & Y_1 & Z_1 & W_1 \\ \hline & M+Q & i(M-Q) & N+P & i(N-P) & M_3+iQ_3 & M_3-iQ_3 & N_3+iP_3 & N_3-iP_3 \\ = & X_2 & Y_2 & Z_2 & W_2 & X_3 & Y_3 & Z_3 & W_3 \end{array}$$

the zero values being

$$\alpha,\ 0,\ \gamma,\ 0\ \ |\ \ \alpha_1,\ 0,\ 0,\ \delta_1\ \ |\ \ \alpha_2,\ 0,\ \gamma_2,\ 0\ \ |\ \ 0,\ \beta_3,\ \gamma_3,\ 0.$$

The equations then are

(Suffixes 0.)				
			X	Z
3		=	α	γ
15	12	=	$-\alpha$	γ
2	1	=	γ	α
14	13	=	$-\gamma$	α
			Y	W
4	7	=	α	$-\gamma$
8	11	=	α	γ
6	5	=	γ	$-\alpha$
10	9	=	γ	α
			$\vartheta^2 0$	
3		=	$\alpha^2+\gamma^2$	
15	12	=	$-(\alpha^2-\gamma^2)$	
2	1	=	$2\alpha\gamma$	

(Suffixes 1.)				
			X	W
3	4	=	$-i\alpha$	$-i\delta$
15	8	=	$i\alpha$	$i\delta$
6	1	=	δ	α
10	13	=	$-\delta$	α
			Y	Z
7		=	α	$-\delta$
11	12	=	α	$-\delta$
2	5	=	$i\delta$	$-i\alpha$
14	9	=	$i\delta$	$i\alpha$
			$\vartheta^2 0$	
3	4	=	$-i(\alpha^2-\delta^2)$	
15	8	=	$i(\alpha^2+\delta^2)$	
6	1	=	$2\alpha\delta$	

(Suffixes 2.)				
			X	Z
15	4	=	$i\alpha$	$i\gamma$
3	8	=	$-i\alpha$	$i\gamma$
14	5	=	$i\gamma$	$-i\alpha$
2	9	=	$-i\gamma$	$-i\alpha$
			Y	W
11		=	α	γ
7	12	=	α	$-\gamma$
10	1	=	γ	α
6	13	=	γ	$-\alpha$
			$\vartheta^2 0$	
15	4	=	$i(\alpha^2-\gamma^2)$	
3	8	=	$-i(\alpha^2+\gamma^2)$	
2	9	=	$-2i\alpha\gamma$	

(Suffixes 3.)				
			Y	Z
15		=	$-\beta$	γ
3	12	=	$-\beta$	γ
10	5	=	γ	$-\beta$
6	9	=	$-\gamma$	$-\beta$
			X	W
11	4	=	β	γ
7	8	=	$-\beta$	$-\gamma$
14	1	=	γ	$-\beta$
2	13	=	γ	β
			$\vartheta^2 0$	
15		=	$-(\beta^2-\gamma^2)$	
3	12	=	$\beta^2+\gamma^2$	
6	9	=	$-2\beta\gamma$.	

115. It will be noticed that the pairs of theta-functions which present themselves in these equations are the same as in the foregoing "Table of the 120 pairs." And the equations show that the four products, each of a pair of theta-functions, belonging to the upper half or to the lower half of any column of the table, are such that any three of the four products are connected by a linear equation. The coefficients of these linear relations are, in fact, functions such as the $\alpha^2+\delta^2$, $\alpha^2-\delta^2$, $2\alpha\delta$ written down at the foot of the several systems of eight equations, and they are consequently products each of two zero-functions c.

Thus (see "The first set, 24 equations") we have

$$
\begin{array}{cccccc|cccccc|cccc}
\multicolumn{6}{c}{\text{(Suffixes 3.)}} & \multicolumn{6}{c}{\text{(Suffixes 3.)}} & \multicolumn{4}{c}{\text{(Suffixes 3.)}} \\
\multicolumn{2}{c}{\vartheta u\cdot\vartheta u} & & X & W & & & & & Y & Z & & \vartheta^2 0 & \vartheta^2 0 & & \\
4 & 8 & = & \alpha & -\delta & & 5 & 9 & = & \alpha & -\delta & & 4 & 8 & = & \alpha^2-\delta^2 \\
 & 12 & = & \alpha & \delta & & 1 & 13 & = & \alpha & \delta & & & 12 & = & \alpha^2+\delta^2 \\
3 & 15 & = & \delta & \alpha & & 2 & 14 & = & \delta & \alpha & & 15 & 3 & = & 2\alpha\delta. \\
7 & 11 & = & -\delta & \alpha & & 6 & 10 & = & -\delta & \alpha & & & & &
\end{array}
$$

116. In the left-hand four of these, omitting successively the first, second, third, and fourth equation, and from the remaining three eliminating the X_3 and W_3, we write down, almost mechanically,

$$
\begin{array}{cc|cccc}
\vartheta u & \vartheta u & & & & \\
\hline
4 & 8 & & +2\alpha\delta, & -\delta^2-\alpha^2, & \alpha^2-\delta^2 \\
 & 12 & -2\alpha\delta, & & -\delta^2+\alpha^2, & \alpha^2+\delta^2 \\
3 & 15 & & & & 2\alpha\delta \\
7 & 11 & -\alpha^2+\delta^2, & \delta^2+\alpha^2, & -2\alpha\delta &
\end{array}
$$

and thence derive the first of the next following system of equations; read

$$
\begin{aligned}
c_4c_{12}\,\vartheta_4\vartheta_8 - c_0c_{12}\,\vartheta_0\vartheta_{12} + c_4c_0\,\vartheta_3\vartheta_{15} + c_4c_8\,\vartheta_7\vartheta_{11} &= 0,\\
-c_4c_{12}\,\vartheta_4\vartheta_8 + c_4c_9\,\vartheta_3\vartheta_{15} - c_3c_{15}\,\vartheta_7\vartheta_{11} &= 0,\\
c_0c_{12}\,\vartheta_4\vartheta_8 + c_4c_9\,\vartheta_0\vartheta_{12} + c_3c_{15}\,\vartheta_7\vartheta_{11} &= 0,
\end{aligned}
$$

where the theta-functions have the arguments $u,\ v$.

Observe that, on writing herein $u=0,\ v=0$, the first three equations become each of them identically $=0$; the fourth equation becomes

$$-c_4c_{12}+c_0c_{12}-c_3c_{15}=0,$$

which is one of the relations between the c's and serves as a verification.

But in the right-hand system, on writing $u=v=0$, each of the four equations becomes identically $=0$.

117. The equations are

ϑ	4.8	0.12	3.15	7.11	= 0,
c		3.15	−0.12	4.8	
	−3.15		−4.8	−0.12	
	0.12	−4.8		3.15	
	−4.8	0.12	−3.15		

ϑ	5.9	1.13	2.14	6.10	= 0,
c		3.15	−0.12	4.8	
	−3.15		4.8	−0.12	
	0.12	−4.8		3.15	
	−4.8	−0.12	−3.15		

ϑ	6.8	2.12	1.15	5.11	= 0,
c		1.15	−2.12	6.8	
	−1.15		6.8	−2.12	
	2.12	−6.8		1.15	
	−6.8	2.12	−1.15		

ϑ	7.9	3.13	0.14	4.10	= 0,
c		1.15	−2.12	6.8	
	−1.15		6.8	−2.12	
	2.12	−6.8		1.15	
	−6.8	2.12	−1.15		

ϑ	0.6	2.4	9.15	11.13	= 0,
c		9.15	−2.4	0.6	
	−9.15		0.6	−2.4	
	2.4	−0.6		9.15	
	−0.6	2.4	−9.15		

ϑ	1.7	3.5	8.14	10.12	= 0,
c		9.15	−2.4	0.6	
	−9.15		0.6	−2.4	
	2.4	−0.6		9.15	
	−0.6	2.4	−9.15		

ϑ	3.6	1.4	9.12	14.11	= 0,
c		9.12	−1.4	3.6	
	−9.12		3.6	−1.4	
	1.4	−3.6		9.12	
	−3.6	1.4	−9.12		

ϑ	2.7	0.5	8.13	10.15	= 0,
c		9.12	−1.4	3.6	
	−9.12		3.6	−1.4	
	1.4	−3.6		9.12	
	−3.6	1.4	−9.12		

ϑ	8.9	0.1	2.3	10.11	= 0,
c		−2.3	0.1	8.9	
	2.3		−8.9	−0.1	
	−0.1	8.9		2.3	
	−8.9	0.1	−2.3		

ϑ	12.13	4.5	6.7	14.15	= 0,
c		−2.3	0.1	8.9	
	2.3		−8.9	−0.1	
	−0.1	8.9		2.3	
	−8.9	0.1	−2.3		

ϑ	4.6	0.2	1.3	5.7	= 0,
c		−1.3	0.2	4.6	
	1.3		−4.6	−0.2	
	−0.2	4.6		1.3	
	−4.6	0.2	−1.3		

ϑ	9.11	13.15	12.14	8.10	= 0,
c		−1.3	0.2	4.6	
	1.3		4.6	−0.2	
	−0.2	−4.6		1.3	
	4.6	0.2	1.3		

ϑ	6.12	2.8	3.9	7.13	= 0,
c		3.9	−2.8	−6.12	
	−3.9		6.12	2.8	
	2.8	−6.12		−3.9	
	6.12	−2.8	3.9		

ϑ	1.11	5.15	4.14	0.10	= 0,
c		3.9	−2.8	6.12	
	−3.9		−6.12	2.8	
	2.8	6.12		−3.9	
	−6.12	−2.8	3.9		

ϑ	6.15		0.9	7.14	= 0,
c		0.9	−1.8	−6.15	
	−0.9		6.15	1.8	
	1.8	−6.15		−0.9	
	6.15	−1.8	0.9		

ϑ	2.11	5.12	4.13	3.10	= 0,
c		0.9	−1.8	6.15	
	−0.9		−6.15	1.8	
	1.8	6.15		−0.9	
	−6.15	−1.8	0.9		

ϑ	4.9	1.12	2.15	7.10	= 0,
c		2.15	−1.12	4.9	
	−2.15		4.9	−1.12	
	1.12	−4.9		2.15	
	−4.9	1.12	−2.15		

ϑ	0.13	5.8	6.11	3.14	= 0,
c		2.15	1.12	−4.9	
	−2.15		−4.9	1.12	
	−1.12	4.9		2.15	
	4.9	−1.12	−2.15		

ϑ	4.12	0.8	1.9	5.13	= 0,
c		−1.9	0.8	4.12	
	1.9		−4.12	−0.8	
	−0.8	4.12		1.9	
	−4.12	0.8	−1.9		

ϑ	3.11	7.15	6.14	2.10	= 0,
c		1.9	−0.8	4.12	
	−1.9		−4.12	0.8	
	0.8	4.12		−1.9	
	−4.12	−0.8	1.9		

ϑ	4.15	3.8	2.9	5.14	$=0$,
c		−2.9	3.8	4.15	
	2.9		−4.15	−3.8	
	−3.8	4.15		2.9	
	−4.15	3.8	−2.9		

ϑ	0.11	7.12	6.13	1.10	$=0$,
c		−2.9	3.8	−4.15	
	2.9		4.15	−3.8	
	−3.8	−4.15		2.9	
	4.15	3.8	−2.9		

ϑ	6.9	3.12	0.15	5.10	$=0$,
c		−0.15	3.12	−6.9	
	0.15		−6.9	3.12	
	−3.12	6.9		−0.15	
	6.9	−3.12	0.15		

ϑ	2.13	7.8	4.11	1.14	$=0$,
c		0.15	3.12	−6.9	
	−0.15		−6.9	3.12	
	−3.12	6.9		0.15	
	6.9	−3.12	−0.15		

ϑ	12.15	0.3	1.2	13.14	$=0$,
c		1.2	−0.3	−12.15	
	−1.2		12.15	0.3	
	0.3	−12.15		−1.2	
	12.15	−0.3	1.2		

ϑ	8.11	4.7	5.6	9.10	$=0$,
c		1.2	−0.3	12.15	
	−1.2		−12.15	0.3	
	0.3	12.15		−1.2	
	−12.15	−0.3	1.2		

ϑ	1.6	3.4	8.15	10.13	$=0$,
c		8.15	−3.4	1.6	
	−8.15		1.6	−3.4	
	3.4	−1.6		8.15	
	−1.6	3.4	−8.15		

ϑ	2.5	0.7	11.12	9.14	$=0$,
c		−8.15	−3.4	1.6	
	8.15		1.6	−3.4	
	3.4	−1.6		−8.15	
	−1.6	3.4	8.15		

ϑ	2.6	0.4	8.12	10.14	$=0$,
c		−8.12	0.4	−2.6	
	8.12		−2.6	0.4	
	−0.4	2.6		−8.12	
	2.6	−0.4	8.12		

ϑ	1.5	3.7	11.15	9.13	$=0$,
c		−8.12	−0.4	2.6	
	8.12		2.6	−0.4	
	0.4	−2.6		−8.12	
	−2.6	0.4	8.12		

118. The foregoing equations may be verified, and it is interesting to verify them, by means of the approximate values of the functions; thus, for one of the equations,

$$c_0c_3\vartheta_3\vartheta_{15}, \quad \text{i.e. } \tfrac{1}{2}\vartheta_3.1.\vartheta_{15}.1,$$

we have

$$\begin{gathered}(2A+2A')(-2A+2A')\\ -1\cdot 1\cdot 2A\cos\tfrac{1}{2}\pi(u+v)+2A'\cos\tfrac{1}{2}\pi(u-v)\cdot\left[-2A\cos\tfrac{1}{2}\pi(u+v)+2A'\cos\tfrac{1}{2}\pi(u-v)\right]\\ +1\cdot 1\cdot -2A\sin\tfrac{1}{2}\pi(u+v)-2A'\sin\tfrac{1}{2}\pi(u-v)\cdot\left[-2A\sin\tfrac{1}{2}\pi(u+v)+2A'\sin\tfrac{1}{2}\pi(u-v)\right]\\ =0\end{gathered}$$

viz. the equation to be verified is here

$$\begin{aligned}&-4A^2+4A'^2\\ &+4A^2\cos^2\tfrac{1}{2}\pi(u+v)-4A'^2\cos^2\tfrac{1}{2}\pi(u-v)\\ &+4A^2\sin^2\tfrac{1}{2}\pi(u+v)-4A'^2\sin^2\tfrac{1}{2}\pi(u-v) \;=\; 0,\end{aligned}$$

which is right.

119. In the equation

$$c_0c_3\vartheta_3\vartheta_{15}, \quad \text{i.e. } 2Q\cdot 1\cdot 2Q\cos\tfrac{1}{2}\pi u\cdot 1-2Q\cdot 1\cdot 2Q\cos\tfrac{1}{2}\pi u\cdot 1=0;$$

this is right, but there is no verification as to the term $c_8c_4\vartheta_7\vartheta_{11}$; taking the more approximate values, the term in question taken negatively, that is, $-c_8c_4\vartheta_7\vartheta_{11}$, is

$$=-(2A+2A')\cdot 2S\cdot -2S\sin\tfrac{1}{2}\pi v\cdot\left[-2A\sin\tfrac{1}{2}\pi(u+v)+2A'\sin\tfrac{1}{2}\pi(u-v)\right],$$

which is

$$=-8S^2(A+A')^2\cos\tfrac{1}{2}\pi u+8S^2(A+A')A\cos\tfrac{1}{2}\pi(u+2v)+8S^2(A+A')A'\cos\tfrac{1}{2}\pi(u-2v),$$

and this ought therefore to be the value of the first two terms, that is, of

$$\begin{aligned}&(2Q+2Q^9-2A-2A')(1-2Q^4-2S^4)\left\{2Q\cos\tfrac{1}{2}\pi u+2Q^9\cos\tfrac{9}{2}\pi u\right.\\ &\qquad\left.+2A\cos\tfrac{1}{2}\pi(u+2v)+2A'\cos\tfrac{1}{2}\pi(u-2v)\right\}(1-2Q^4\cos\pi u+2S^4\cos\pi v)\\ &-(2Q+2Q^9+2A+2A')(1-2Q^4+2S^4)\left\{2Q\cos\tfrac{1}{2}\pi u+2Q^9\cos\tfrac{9}{2}\pi u\right.\\ &\qquad\left.-2A\cos\tfrac{1}{2}\pi(u+2v)-2A'\cos\tfrac{1}{2}\pi(u-2v)\right\}(1-2Q^4\cos\pi u-2S^4\cos\pi v),\end{aligned}$$

which to the proper degree of approximation is

$$\begin{aligned}&=(2Q-4Q^5-4QS^4+2Q^9-2A-2A')\left[2Q\cos\tfrac{1}{2}\pi u-4Q^5\cos\tfrac{1}{2}\pi u\cos\pi u\right.\\ &\qquad\left.+4QS^4\cos\tfrac{1}{2}\pi u\cos\pi v+2Q^9\cos\tfrac{9}{2}\pi u+2A\cos\tfrac{1}{2}\pi(u+2v)+2A'\cos\tfrac{1}{2}\pi(u-2v)\right]\\ &\quad-(2Q-4Q^5+4QS^4+2Q^9+2A+2A')\left[2Q\cos\tfrac{1}{2}\pi u-4Q^5\cos\tfrac{1}{2}\pi u\cos\pi u\right.\\ &\qquad\left.-4QS^4\cos\tfrac{1}{2}\pi u\cos\pi v+2Q^9\cos\tfrac{9}{2}\pi u-2A\cos\tfrac{1}{2}\pi(u+2v)-2A'\cos\tfrac{1}{2}\pi(u-2v)\right].\end{aligned}$$

This is

$$(2M_0 - 2\Omega_0)(2M + 2\Omega) - (2M_0 + 2\Omega_0)(2M - 2\Omega) = 8(M_0\Omega - M\Omega_0),$$

if for a moment

$$M = Q\cos\tfrac{1}{2}\pi u - 2Q^5\cos\tfrac{1}{2}\pi u\cos\pi u + Q^9\cos\tfrac{9}{2}\pi u, \quad M_0 = Q - 2Q^5 + Q^9,$$

$$\Omega = 2QS^4\cos\tfrac{1}{2}\pi u\cos\pi v + A\cos\tfrac{1}{2}\pi(u+2v) + A'\cos\tfrac{1}{2}\pi(u-2v), \quad \Omega_0 = 2QS^4 + A + A',$$

or substituting and reducing, the value of $8(M_0\Omega - M\Omega_0)$ to the proper degree of approximation is found to be

$$= -8Q(2QS^4 + A + A')\cos\tfrac{1}{2}\pi u + 8(Q^2S^4 + 8QA)\cos\tfrac{1}{2}\pi(u+2v) + 8(Q^2S^4 + 8QA')\cos\tfrac{1}{2}\pi(u-2v),$$

which in virtue of the relations $QA = A'S^4$, $QA' = A'S^4$; $Q^2S^4 = AA'$, is equal to the foregoing value of $c_8c_4\vartheta_7\vartheta_{11}$. I have thought it worth while to give this somewhat elaborate verification.

Résumé of the foregoing results.

120. In what precedes we have all the quadric relations between the 16 double theta-functions; we have the linear relations between squares (squared functions) or say between pairs (products of two functions): the number of the asyzygetic linear relations between squares is obviously = 12, and that of the asyzygetic linear relations between pairs is = 60 (since each of the 30 tetrads of pairs gives two asyzygetic relations): there are thus in all 12 + 60, = 72, asyzygetic linear relations. But these constitute only a 13-fold relation between the functions, viz. they are such as to give for the ratios of the 16 functions expressions depending upon two arbitrary parameters, x, y. Or taking the 16 functions as the coordinates of a point in 15-dimensional space, these coordinates are connected by a 13-fold relation (expressed by means of the foregoing system of 72 quadric equations), and the locus is thus a 13-fold, or two-dimensional, locus in 15-dimensional space.

Hence, taking any four of the functions, these are connected by a single equation; that is, regarding the four functions as the coordinates of a point in ordinary space, the locus of the point is a surface.

In particular, the four functions may be any four functions belonging to a hexad: by what precedes there is then a linear relation between the squares of the four functions: or the locus is a quadric surface. Each hexad gives 15 such surfaces, or the number of quadric surfaces is $(16 \times 15 =)240$.

The 16-nodal quartic surfaces.

121. If the four functions are those contained in any two pairs out of a tetrad of pairs (see the foregoing "Table of the 120 pairs"), then the locus is a quartic surface, which is, in fact, a Kummer's 16-nodal quartic surface. For if for a moment $x.y$ and $z.w$ are two pairs out of a tetrad, and $r.s$ be either of the remaining pairs of the tetrad; then we have rs a linear function of xy and zw; squaring, r^2s^2 is a linear function of x^2y^2, $xyzw$, z^2w^2; but we then have r^2 and s^2 each of them a linear function of x^2, y^2, z^2, w^2; or substituting we have an equation of the fourth order, containing terms of the second order in (x^2, y^2, z^2, w^2), and also a term in $xyzw$. It is clear that, if instead of $r.s$ we had taken the remaining pair of the tetrad, we should have obtained the same quartic equation in (x, y, z, w). And moreover it appears by inspection that, if xy and zw are pairs in a tetrad, then xz and yw are pairs in a second tetrad, and xw and yz are pairs in a third tetrad: we obtain in each case the same quartic equation. We have from each tetrad of pairs six sets of four functions (x, y, z, w): and the number of such sets is thus $({}_6C_2 \cdot 30 =)$ 60; these are shown in the foregoing "Table of the 60 Göpel tetrads," viz. taking as coordinates of a point the four functions in any tetrad of this table, the locus is a 16-nodal quartic surface.

122. To exhibit the process I take a tetrad 4, 7, 8, 11 containing two odd functions; and representing these for convenience by x, y, z, w, viz. writing

$$\vartheta_4, \vartheta_7, \vartheta_8, \vartheta_{11}(u) = x, y, z, w,$$

we have then X, Y, Z, W linear functions of the four squares, viz. it is easy to obtain

$$\begin{aligned} \alpha(x^2+z^2) - \beta(y^2+w^2) &= 2(\alpha^2-\beta^2)\,X, \\ \beta(\ \ ,,\ \ ) - \alpha(\ \ ,,\ \ ) &= 2(\ \ ,,\ \ )\,W, \\ -\beta(x^2-z^2) + \gamma(y^2-w^2) &= 2(\beta^2-\gamma^2)\,Y, \\ -\gamma(\ \ ,,\ \ ) + \beta(\ \ ,,\ \ ) &= 2(\ \ ,,\ \ )\,Z. \end{aligned}$$

Also considering two other functions $\vartheta_0(u)$ and $\vartheta_{12}(u)$, or as for shortness I write them, ϑ_0 and ϑ_{12}, we have

$$\vartheta_0^2 = \alpha X + \beta Y + \gamma Z + \delta W,$$

$$\vartheta_{12}^2 = \alpha X - \beta Y - \gamma Z + \delta W,$$

and substituting the foregoing values of X, Y, Z, W, we find

$$M\vartheta_0^2 = Ax^2 + By^2 + Cz^2 + Dw^2,$$

$$M\vartheta_{12}^2 = Cx^2 + Dy^2 + Az^2 + Bw^2,$$

where, writing down the values first in terms of $\alpha, \beta, \gamma, \delta$ and then in terms of the c's, we have

$$\begin{aligned} M &= (\alpha^2-\beta^2)(\beta^2-\gamma^2) &&= \tfrac{1}{4}\cdot c_4^2 - c_{12}^2, \\ A &= \alpha\delta\beta^2 - \alpha^2\beta\gamma &&= \alpha\, c_3\, c_4^2, \\ B &= -\alpha\delta(\beta^2-\gamma^2) + \beta\gamma(\alpha^2-\delta^2) &&= c_4^2\, c_{12}^2 - c_3 c_{15}, \\ C &= \alpha^2\beta^2 - \gamma^2\delta^2 &&= c_4 c_{12}, \end{aligned}$$

and we then have further

$$c_0 c_3 \vartheta_0 \vartheta_{12} = c_4 c_{12} xz + c_3 c_{15} yw,$$

that is,

$$M\vartheta_0\vartheta_{12} = \sqrt{a}\, xz + \sqrt{b}\, yw,$$

whence equating the two values of $\vartheta_0^2\vartheta_{12}^2$ we have the required quartic equation in x, y, z, w.

123. But the reduction is effected more simply if instead of the c's we introduce the rectangular coefficients a, b, c, &c. We then have

$$M = (c''^2 - b'^2), \quad A = -a''c, \quad C = a'b,$$

$$B = -b'c' - b''c'' = bc, \quad D = b'b'' + c'c'' = a'a'';$$

and the equations become

$$\begin{aligned}
(c''^2 - b'^2)\vartheta_0^2 &= -a''cx^2 + bcy^2 + a'bz^2 - a'a''w^2,\\
(c''^2 - b'^2)\vartheta_{12}^2 &= a'bx^2 - a'a''y^2 - a''cz^2 + bcw^2,\\
\sqrt{b'^2c''^2}\,\vartheta_0\vartheta_{12} &= \sqrt{a}\, xz + \sqrt{b}\, yw,
\end{aligned}$$

so that the elimination gives

$$b'c''(-x^2c + bcy^2 + a'bz^2 - a'a''w^2)(a'bx^2 - a'a''y^2 - a''cz^2 + bcw^2) = (c''^2 - b'^2)^2\big\{ax^2z^2 - b''c'y^2w^2 + 2\sqrt{-ab''c'}\, xyzw\big\},$$

viz. this is

$$\begin{aligned}
&- a'a''b'cc''(x^4 + y^4 + z^4 + w^4)\\
&+ a'b'cc''(a''^2 + b^2)(x^2z^2 + y^2w^2)\\
&+ \big[b'c''(a''^2b^2 + a'^2c^2) - a(b'^2 - c''^2)^2\big]x^2z^2\\
&+ \big[b'c''(a'^2a''^2 + b^2c^2) + b''c'(b'^2 - c''^2)^2\big]y^2w^2\\
&- a''bb'c''(a'^2 + c^2)(x^2w^2 - y^2z^2)\\
&- 2(b'^2 - c''^2)^2\sqrt{-ab''c'}\, xyzw = 0.
\end{aligned}$$

124. In this equation the coefficients of x^2z^2 and y^2w^2 are each $= a'a''bc(b'^2 + c''^2)$, as at once appears from the identities

$$\begin{aligned}
a'b \cdot b' - c'' \cdot a''c &= a(b'^2 - c''^2),\\
a'b \cdot c'' - b' \cdot a''c &= (b'^2 - c''^2),\\
a'a'' \cdot b' - c'' \cdot bc &= -b''(b'^2 - c''^2),\\
a'a'' \cdot c - b' \cdot bc &= c'(b'^2 - c''^2),
\end{aligned}$$

by multiplying together in each pair the left-hand and the right-hand sides respectively. Substituting and dividing by $-a'a''bb'cc''$, we have

$$\frac{x^4 + y^4 + z^4 + w^4}{a''^2 + b^2} - \frac{b'^2 + c''^2}{} + \frac{a'^2 + c^2}{} \quad \text{[arrangement as in scan]}$$

$$+ \frac{2(b'^2 - c''^2)^2\sqrt{-ab''c'}}{a'a''bb'cc''}\, xyzw = 0.$$

or, if we herein restore the c's in place of the rectangular coefficients, this is

$$\frac{x^4+y^4+z^4+w^4}{c_4^2c_{12}^2}-\frac{c_3^2+c_{15}^2}{c_4^2c_{12}^2}(x^2z^2+y^2w^2)-\frac{c_3^2c_{15}^2}{c_4^2c_{12}^2}(x^2w^2+y^2z^2)+\frac{2(c_4^2-c_{12}^2)^2\,c_0c_3}{c_4c_{12}\cdot c_0c_3}\,xyzw=0,$$

which is the equation of the 16-nodal quartic surface.

Substituting for x, y, z, w their values $\vartheta_4, \vartheta_7, \vartheta_8, \vartheta_{11}(u)$, we have the equation connecting the four theta-functions 4, 7, 8, 11 of a Göpel tetrad. And there is an equation of the like form between the four functions of any other Göpel tetrad: for obtaining the actual equations some further investigation would be necessary.

The xy-expressions of the theta-functions.

125. The various quadric relations between the theta-functions, admitting that they constitute a 13-fold relation, show that the theta-functions may be expressed as proportional to functions of two arbitrary parameters x, y; and two of these functions being assumed at pleasure the others of them would be determinate: we have of course (though it would not be easy to arrive at it in this manner) such a system in the foregoing expressions of the 16 functions in terms of x, y; and conversely these expressions must satisfy identically the quadric relations between the theta-functions.

126. To show that this is so as to the general form of the equations, consider first the xy-factors $\sqrt{a}$, $\sqrt{ab}$, &c. As regards the squared functions $(\sqrt{ab})^2$, we have for instance

$$(\sqrt{ab})^2=\tfrac{1}{\theta^2}\{ab\,f_2c_2d_2e_2+a_2b_2f_2cde+2\sqrt{XY}\},$$

$$(\sqrt{cd})^2=\tfrac{1}{\theta^2}\{cd\,f_2a_2b_2e_2+c_2d_2f_2abe+2\sqrt{XY}\},$$

each of these contains the same irrational part $\frac{2}{\theta^2}\sqrt{XY}$, and the difference is therefore rational: and it is moreover integral, for we have

$$(\sqrt{ab})^2-(\sqrt{cd})^2=\tfrac{1}{\theta^2}(abc_2d_2-a_2b_2cd)(fe_2-f_2e),$$

where each factor divides by θ, and consequently the product by θ^2; the value is in fact

$$=(e-f)\begin{vmatrix}1, & x+y, & xy\\ 1, & a+b, & ab\\ 1, & c+d, & cd\end{vmatrix}$$

a linear function of 1, $x+y$, xy. This is the case as regards the difference of any two of the squares $(\sqrt{ab})^2$, $(\sqrt{ac})^2$, &c.; hence selecting any one of these squares, for instance $(\sqrt{de})^2$, any other of the squares is of the form

$$\lambda + \mu(x+y) + \nu xy + \rho(\sqrt{de})^2, \qquad (\rho = 1);$$

and obviously, the other squares $(\sqrt{a})^2$, &c., are of the like form, the last coefficient ρ being $= 0$. We hence have the theorem that each square can be expressed as a linear function of any four (properly selected) squares.

127. But we have also the theorem of the 16 Kummer hexads.

Obviously the six squares

$$(\sqrt{a})^2,\ (\sqrt{b})^2,\ (\sqrt{c})^2,\ (\sqrt{d})^2,\ (\sqrt{e})^2,\ (\sqrt{f})^2$$

are a hexad, viz. each of these is a linear function of 1, $x+y$, xy: and therefore selecting any three of them, each of the remaining three can be expressed as a linear function of these.

But further the squares $(\sqrt{a})^2$, $(\sqrt{b})^2$, $(\sqrt{ab})^2$, $(\sqrt{cd})^2$, $(\sqrt{ce})^2$, $(\sqrt{de})^2$ form a hexad.

For reverting to the expression obtained for $(\sqrt{ab})^2 - (\sqrt{cd})^2$, the determinant contained therein is a linear function of aa_2 and bb_2, that is, of $(\sqrt{a})^2$ and $(\sqrt{b})^2$; we, in fact, have

$$(a-b)\begin{vmatrix} 1, & x+y, & xy \\ 1, & a+b, & ab \\ 1, & c+d, & cd \end{vmatrix} = (b-c)(b-d)(a-x)(a-y) - (a-c)(a-d)(b-x)(b-y).$$

Hence $(\sqrt{ab})^2 - (\sqrt{cd})^2$ is a linear function of $(\sqrt{a})^2$, $(\sqrt{b})^2$; and by a mere interchange of letters $(\sqrt{ab})^2 - (\sqrt{ce})^2$, $(\sqrt{ab})^2 - (\sqrt{de})^2$, are each of them also a linear function of $(\sqrt{a})^2$ and $(\sqrt{b})^2$; whence the theorem. And we have thus all the remaining 15 hexads.

128. We have a like theory as regards the products of pairs of functions. A tetrad of pairs is of one of the two forms

$$\sqrt{a}\sqrt{b},\ \sqrt{ac}\sqrt{bc},\ \sqrt{ad}\sqrt{bd},\ \sqrt{ae}\sqrt{be}, \quad \text{and} \quad \sqrt{f}\sqrt{ab},\ \sqrt{c}\sqrt{de},\ \sqrt{d}\sqrt{ce},\ \sqrt{e}\sqrt{cd};$$

in the first case the terms are

$$\begin{gathered} \sqrt{aa_2bb_2}, \\ \tfrac{1}{\theta}\{(ab_2+a_2b)\sqrt{cdef\,c_2d_2e_2f_2} + (cfd_2e_2+c_2f_2de)\sqrt{aa_2bb_2}\}, \\ \tfrac{1}{\theta}\{\ \ ,,\ \ +(dfc_2e_2+d_2f_2ce)\sqrt{aa_2bb_2}\}, \\ \tfrac{1}{\theta}\{\ \ ,,\ \ +(efc_2d_2+e_2f_2cd)\sqrt{aa_2bb_2}\}, \end{gathered}$$

and as regards the last three terms the difference of any two of them is a mere constant multiple of $\sqrt{aa_2bb_2}$; for instance, the second term $-$ the third term is

$$\tfrac{1}{\theta}(cd_2-c_2d)(fe_2-f_2e)\sqrt{aa_2bb_2} = (c-d)(f-e)\sqrt{aa_2bb_2}.$$

we have thus a tetrad such that, selecting any two terms, each of the remaining terms is a linear function of these.

In the second case, the terms are

$$\tfrac{1}{\theta}\{f\sqrt{abc_2d_2e_2f_2}+f_2\sqrt{a_2b_2cdef}\},$$

$$\begin{array}{l}\tfrac{1}{\theta}\{c \quad ,, \quad +c_2 \quad ,, \quad \},\\ \tfrac{1}{\theta}\{d \quad ,, \quad +d_2 \quad ,, \quad \},\\ \tfrac{1}{\theta}\{e \quad ,, \quad +e_2 \quad ,, \quad \},\end{array}$$

whence clearly the four terms are a tetrad as above. And it may be added that any linear function of the four terms is of the form

$$\tfrac{1}{\theta}\{(\lambda+\mu x)\sqrt{abc_2d_2e_2f_2}+(\lambda+\mu y)\sqrt{a_2b_2cdef}\}.$$

129. Considering next the actual equations between the squared theta-functions, take as a specimen

$$c_2^4(\sqrt{ab})^2-c_4^4(\sqrt{cd})^2+c_5^4(\sqrt{ce})^2-c_6^4(\sqrt{de})^2=0,$$

that is,

$$c_6^4(\sqrt{ab})^2-c_3^4(\sqrt{cd})^2+c_4^4(\sqrt{ce})^2-c_5^4(\sqrt{de})^2=0,$$

where $c_8, c_3, c_4, c_5=\sqrt{ab}, \sqrt{cd}, \sqrt{ce}, \sqrt{de}$ respectively. Since the functions $(\sqrt{ab})^2$, &c., contain the same irrational term $\frac{2}{\theta^2}\sqrt{XY}$, it is clear that the equation can only be true if

$$c_8^4-c_3^4+c_4^4-c_5^4=0;$$

and, this being so, it will be true if

$$c_3^4\{(\sqrt{ab})^2-(\sqrt{cd})^2\}-c_4^4\{(\sqrt{ab})^2-(\sqrt{ce})^2\}+c_5^4\{(\sqrt{ab})^2-(\sqrt{de})^2\}=0,$$

where, by what precedes, each of the terms in { } is a linear function of $(\sqrt{a})^2$ and $(\sqrt{b})^2$. Attending first to the term in $(\sqrt{a})^2$, the coefficient hereof is

$$ef\cdot bc\cdot bd\cdot c_3^4-df\cdot bc\cdot be\cdot c_4^4+cf\cdot bd\cdot be\cdot c_5^4,$$

where for shortness bc, bd, &c., are written to denote the differences $b-c,\ b-d$, &c.

substituting for c_3^4 its value $(\sqrt{cd})^4=cd\cdot cf\cdot df\cdot ab\cdot ae\cdot be$, and similarly for c_4^4 and c_5^4 their values, $=ce\cdot cf\cdot ef\cdot ab\cdot ad\cdot bd$, and $de\cdot df\cdot ef\cdot ab\cdot ac\cdot bc$ respectively, the whole expression contains the factor $ab\cdot bc\cdot bd\cdot be\cdot cf\cdot df\cdot ef$, and throwing this out, the equation to be verified becomes

$$cd\cdot ae-ce\cdot ad+de\cdot ac=0,$$

which is true identically. The verification is thus made to depend upon that of $c_3^4 - c_4^4 + c_5^4 - c_8^4 = 0$; and similarly for the other relations between the squared functions, the verification depends upon relations containing the fourth powers, or the products of squares, of the constants c and k.

130. Among these are included the before-mentioned system of equations involving the fourth powers or the products of squares of only the constants c; and it is interesting to show how these are satisfied identically by the values $c_0 = \sqrt{bd}$, &c.

Thus one of these equations is $c_{11}^4 + c_3^4 + c_5^4 = c_0^4$; substituting the values, there is a factor ce which divides out, and the resulting equation is

$$ad \cdot af \cdot df \cdot bc \cdot be + cf \cdot ef \cdot ab \cdot ad \cdot bd + ab \cdot af \cdot bf \cdot cd \cdot de - ac \cdot ae \cdot bd \cdot bf \cdot df = 0.$$

There are here terms in a^3, a^2, a which should separately vanish; for the terms in a^3, the equation becomes

$$df \cdot bc \cdot be + bd \cdot cf \cdot ef + bf \cdot cd \cdot de - bd \cdot bf \cdot df = 0,$$

which is easily verified; and the equations in a^2 and a^0 may also be verified.

An equation involving products of the squares is $c_8^2c_9^2 - c_4^2c_5^2 + c_1^2c_2^2 = 0$. The term $c_8^2c_9^2$ is here $\sqrt{adf \cdot bce}\sqrt{def \cdot abc}$ which is $= \sqrt{(bc)^2(df)^2 \cdot ab \cdot ac \cdot ad \cdot af \cdot be \cdot ce \cdot de \cdot ef}$, which is taken $= bc \cdot df\sqrt{ab \cdot ac \cdot ad \cdot af \cdot be \cdot ce \cdot de \cdot ef}$; similarly the values of $c_4^2c_5^2$ and $c_1^2c_2^2$ are $= bd \cdot cf$ and $bf \cdot cd$ each multiplied by the same radical, and the equation to be verified is

$$bc \cdot df - bd \cdot cf + bf \cdot cd = 0,$$

which is right: the other equations may be verified in a similar manner.

131. Coming next to the equations connecting the pairs of theta-functions, for instance

$$c_8c_{12}c_0c_{12}\left[\sqrt{bd}\,\sqrt{ad} - \sqrt{be}\,\sqrt{ae}\right] + c_4c_5\kappa_9\kappa_{11} \cdot \sqrt{b}\sqrt{a} = 0,$$

this is

$$c_8c_{12}c_0c_{12}\left[\sqrt{bd}\,\sqrt{ad} - \sqrt{be}\,\sqrt{ae}\right] + c_4c_5\kappa_9\kappa_{11} \cdot \sqrt{b}\sqrt{a} = 0,$$

the products $\sqrt{bd}\,\sqrt{ad}$ and $\sqrt{be}\,\sqrt{ae}$ contain besides a common term the terms

$$\tfrac{1}{\theta}(dfc_2e_2 + d_2f_2ce)\sqrt{aa_2bb_2}, \quad \text{and} \quad \tfrac{1}{\theta}(efc_2d_2 + e_2f_2cd)\sqrt{aa_2bb_2};$$

hence their difference contains $\frac{1}{\theta}(de_2 - d_2e)(fc_2 - f_2c)\sqrt{aa_2bb_2}$ which is $= de \cdot fc\sqrt{aa_2bb_2}$, that is, $de \cdot fc\sqrt{a}\sqrt{b}$: hence the equation to be verified is

$$de \cdot fc \cdot c_8c_{12}c_0c_{12} + c_4c_5\kappa_9\kappa_{11} = 0;$$

$c_8c_{12}c_0c_{12} = \sqrt{bef \cdot acd}\sqrt{adf \cdot bcd}\sqrt{bdf \cdot ace}\sqrt{adf \cdot bce}$, where under the fourth root we have 24 factors, which are, in fact, 12 factors twice repeated; and if we write

$$\Pi = ab \cdot ac \cdot ad \cdot ae \cdot af \cdot bc \cdot bd \cdot be \cdot bf \cdot cd \cdot ce \cdot cf \cdot de \cdot df \cdot ef$$

for the product of all the 15 factors, then the 12 factors are in fact all those of Π, except ab, cf, de; viz. we have

$$c_8c_{12}c_0c_{12} = \sqrt{\Pi^2/(ab)^2(cf)^2(de)^2}.$$

Again, $c_4c_5\kappa_9\kappa_{11} = \sqrt{acf\cdot bde}\,\sqrt{bcf\cdot ade}\,\sqrt{acdef}\,\sqrt{bcdef}$, is a fourth root of a product of 32 factors, which are in fact 16 factors twice repeated, and in the 16 factors, ab does not occur, cf and de occur each twice, and the other 12 factors each once: we thus have

$$c_4c_5\kappa_9\kappa_{11} = \sqrt{\Pi^2(cf)^2(de)^2 \div (ab)^2},$$

and the relation to be verified assumes the form

$$fc\cdot de\sqrt{\Pi^2/(cf)^2(de)^2} + \sqrt{\Pi^2(cf)^2(de)^2} = 0,$$

which, taking $fc\cdot de = -\sqrt{(cf)^2(de)^2}$, is right. And so for the other equations. It will be observed that, in the equation $de\cdot fc\cdot c_8c_{12}c_0c_{12} + c_4c_5\kappa_9\kappa_{11} = 0$, and in the other equations upon which the verifications depend, there is no ambiguity of sign: the signs of the radicals have to be determined consistently with all the equations which connect the c's and the k's; that this is possible appears evident *à priori*, but the actual verification presents some difficulty. I do not here enter further into the question.

Further results of the product-theorem, the $u \pm u'$ formulæ.

132. Recurring now to the equations in $u+u'$, $u-u'$, by putting therein $u'=0$, we can express X, Y, Z, W in terms of four of the squared functions of u, and by putting $u=0$ we can express X', Y', Z', W' in terms of four of the squared functions of u'; and, substituting in the original equations, we have the products

$$\vartheta(\;)_{u+u'}\cdot\vartheta(\;)_{u-u'}$$

in terms of the squared functions of u and u'.

Selecting as in a former investigation the functions 4, 7, 8, 11, which were called x, y, z, w, it is more convenient to use single letters to represent the squared functions. I write

$$\begin{array}{cccc}
\vartheta_4(u+u')\,\vartheta_4(u-u') & \vartheta_4^2u & \vartheta_4^2u' & \vartheta_4^2 0\\
4\quad 4 = P, & 4 = p, & 4 = p', & 4 = p_0\,(=c_4^2),\\
7\quad 7 = Q, & 7 = q, & 7 = q', & 7 = 0,\\
8\quad 8 = R, & 8 = r, & 8 = r', & 8 = r_0\,(=c_8^2),\\
11\quad 11 = S, & 11 = s, & 11 = s', & 11 = 0.
\end{array}$$

Then

$$\begin{array}{lll}
\quad X\;Y\;Z\;W & \quad X\;Y\;Z\;W & \quad X'\;Y'\;Z'\;W'\\
P = X^2 - Y^2 + Z^2 - W^2, & p = \alpha - \beta + \gamma - \delta, & p' = \alpha - \beta + \gamma - \delta,\\
Q = W^2 - Z^2 + Y^2 - X^2, & q = \delta - \gamma + \beta - \alpha, & q' = \delta - \gamma + \beta - \alpha,\\
R = X^2 + Y^2 - Z^2 - W^2, & r = \alpha + \beta - \gamma - \delta, & r' = \alpha + \beta - \gamma - \delta,\\
S = W^2 + Z^2 - Y^2 - X^2, & s = \delta + \gamma - \beta - \alpha, & s' = \delta + \gamma - \beta - \alpha.
\end{array}$$

Hence

$$\begin{array}{ll}
\alpha(p+r) - \delta(q+s) = 2(\alpha^2-\delta^2)X, & \alpha(p'+r') - \delta(q'+s') = 2(\alpha^2-\delta^2)X',\\
\delta\quad ,,-\alpha\quad ,, = 2\quad ,,\; W, & \delta\quad ,,-\alpha\quad ,, = 2\quad ,,\; W',\\
-\beta(p-r) + \gamma(q-s) = 2(\beta^2-\gamma^2)Y, & -\beta(p'-r') + \gamma(q'-s') = 2(\beta^2-\gamma^2)Y',\\
-\gamma\quad ,,+\beta\quad ,, = 2\quad ,,\; Z, & -\gamma\quad ,,+\beta\quad ,, = 2\quad ,,\; Z'.
\end{array}$$

By means of these values, we have

$$4(\alpha^2-\delta^2)^2X'X = \alpha^2(p+r)(p'+r') + \delta^2(q+s)(q'+s') - \alpha\delta\big[(p+r)(q'+s') + (p'+r')(q+s)\big],$$
$$4\quad ,,\quad W'W = \delta^2\quad ,,+\alpha^2\quad ,,-\alpha\delta\big[\quad ,,\quad\big],$$
$$4(\beta^2-\gamma^2)^2Y'Y = \beta^2(p-r)(p'-r') + \gamma^2(q-s)(q'-s') - \beta\gamma\big[(p-r)(q'-s') + (p'-r')(q-s)\big],$$
$$4\quad ,,\quad Z'Z = \gamma^2\quad ,,+\beta^2\quad ,,-\beta\gamma\big[\quad ,,\quad\big].$$

Hence

$$4(\alpha^2-\delta^2)(X'X-W'W) = (p+r)(p'+r') - (q+s)(q'+s'),$$
$$4(\beta^2-\gamma^2)(Y'Y-Z'Z) = (p-r)(p'-r') - (q-s)(q'-s'),$$

and substituting in the expressions for P and R,

$$4(\alpha^2-\delta^2)(\beta^2-\gamma^2)P = (\beta^2-\gamma^2)\big[(p+r)(p'+r')-(q+s)(q'+s')\big] - (\alpha^2-\delta^2)\big[(p-r)(p'-r')-(q-s)(q'-s')\big],$$
$$4\quad ,,\quad R = \quad ,,-\quad ,,.$$

Similarly

$$4(\alpha^2-\delta^2)^2W'X = \alpha\delta\big[(p+r)(p'+r') + (q+s)(q'+s')\big] - \alpha^2(p+r)(q'+s') - \delta^2(q+s)(p'+r'),$$
$$4\quad ,,\quad X'W = \quad ,,-\delta^2\quad ,,-\alpha^2\quad ,,,$$
$$4(\beta^2-\gamma^2)^2Z'Y = \beta\gamma\big[(p-r)(p'-r') + (q-s)(q'-s')\big] - \beta^2(p-r)(q'-s') - \gamma^2(q-s)(p'-r'),$$
$$4\quad ,,\quad Y'Z = \quad ,,-\gamma^2\quad ,,-\beta^2\quad ,,.$$

whence

$$4(\alpha^2-\delta^2)(W'X-X'W) = -\big[(p+r)(q'+s') - (p'+r')(q+s)\big],$$
$$4(\beta^2-\gamma^2)(Z'Y-Y'Z) = -\big[(p-r)(q'-s') - (p'-r')(q-s)\big],$$

and substituting in the expressions for Q and S

$$4(\alpha^2-\delta^2)(\beta^2-\gamma^2)Q = -(\beta^2-\gamma^2)\big[(p+r)(q'+s')-(p'+r')(q+s)\big] + (\alpha^2-\delta^2)\big[(p-r)(q'-s')-(p'-r')(q-s)\big],$$
$$,,\quad S = -\quad ,,-\quad ,,.$$

133. Hence, collecting and reducing,

$$4(\alpha^2-\delta^2)(\beta^2-\gamma^2)P = -(\alpha^2-\beta^2+\gamma^2-\delta^2)(pp'-qq'+rr'-ss')+(\alpha^2+\beta^2-\gamma^2-\delta^2)(pr'+p'r-qs'-q's),$$
$$4 \quad ,, \quad R = ,,$$
$$4 \quad ,, \quad Q = ,,$$
$$4 \quad ,, \quad S = ,,$$

we have

$$p_0\,(=c_4^2) = \alpha^2-\beta^2+\gamma^2-\delta^2, \qquad r_0\,(=c_8^2) = \alpha^2+\beta^2-\gamma^2-\delta^2,$$

and thence

$$r_0^2-p_0^2 = 4(\alpha^2-\delta^2)(\beta^2-\gamma^2);$$

the equations hence become

$$(r_0^2-p_0^2)P = -p_0(pp'-qq'+rr'-ss')+r_0(pr'+p'r-qs'-q's),$$
$$,, \quad R = r_0(\ \ ,,\ \ ) - p_0(\ \ ,,\ \ ),$$
$$,, \quad Q = p_0(pq'-p'q+rs'-r's) - r_0(\ \ ,,\ \ ),$$
$$,, \quad S = -r_0(\ \ ,,\ \ ) + p_0(\ \ ,,\ \ ).$$

On writing in the equations $u'=0$, then $P, Q, R, S, p', q', r', s'$ become $= p, q, r, s, p_0, 0, r_0, 0$; and the equations are (as they should be) true identically. The equations may be written

$$(8-4)\vartheta\tfrac{u+u'}{4}\vartheta\tfrac{u-u'}{4} = -4(\vartheta_4u\vartheta_4u'-\vartheta_7u\vartheta_7u'+\vartheta_8u\vartheta_8u'-\vartheta_{11}u\vartheta_{11}u')+8(\vartheta_4u\vartheta_8u'+\vartheta_8u\vartheta_4u'-\vartheta_7u\vartheta_{11}u'-\vartheta_{11}u\vartheta_7u'),$$
$$(\ \ ,,\ \ )\vartheta_8\vartheta_8 = +8(\ \ ,,\ \ )-4(\ \ ,,\ \ ),$$
$$(\ \ ,,\ \ )\vartheta_7\vartheta_7 = +4(\vartheta_4u\vartheta_7u'-\vartheta_7u\vartheta_4u'+\vartheta_8u\vartheta_{11}u'-\vartheta_{11}u\vartheta_8u')-8(\vartheta_4u\vartheta_{11}u'-\vartheta_{11}u\vartheta_4u'+\vartheta_8u\vartheta_7u'-\vartheta_7u\vartheta_8u'),$$
$$(\ \ ,,\ \ )\vartheta_{11}\vartheta_{11} = -8(\ \ ,,\ \ )+4(\ \ ,,\ \ ).$$

There is of course such a system for each of the 60 Göpel tetrads.

Differential relations connecting the theta-functions with the quotient-functions.

134. Imagine p, q, r, s, &c., changed into x'^2, y'^2, z'^2, w'^2, &c.; that is, let x, y, z, w represent the theta-functions 4, 7, 8, 11 of u, v; and similarly x', y', z', w' those of u', v', and $x_0, 0, z_0, 0$ those of 0, 0. Let u', v' be each of them indefinitely small; and take $\partial, = u'\dfrac{d}{du}+v'\dfrac{d}{dv}$, as the symbol of total differentiation in regard to u, v, the infinitesimals u' and v' being arbitrary: then, as far as the second order, we have in general

$$\vartheta(u+u', v+v') = \vartheta(u,v)+\partial\vartheta(u,v)+\tfrac{1}{2}\partial^2\vartheta(u,v),$$

and hence

$$P = (x + \partial x + \tfrac{1}{2}\partial^2 x)(x - \partial x + \tfrac{1}{2}\partial^2 x), \quad = x^2 + \{x\partial^2 x - (\partial x)^2\},$$

and similarly for Q, R, S. Moreover, observing that x' and z' are even functions, y' and w' are odd functions, of u', v', we have

$$x',\ y',\ z',\ w' = x_0 + \tfrac{1}{2}\partial^2 x_0,\ \partial y_0,\ z_0 + \tfrac{1}{2}\partial^2 z_0,\ \partial w_0,$$

where $\partial^2 x_0$, ∂y_0, &c., are what $\partial^2 x$, ∂y, &c., become on writing therein $u = 0$, $v = 0$; $\partial y_0, \partial w_0$ are of course linear functions, $\partial^2 x_0, \partial^2 z_0$ quadric functions of u' and v'. The values of x'^2, y'^2, z'^2, w'^2 are thus $x_0^2 + x_0\partial^2 x_0$, $(\partial y_0)^2$, $z_0^2 + z_0\partial^2 z_0$, $(\partial w_0)^2$; and we have

$$\begin{array}{r} x_0^2 x_0\partial^2 x_0 \quad (\partial y_0)^2 \quad z_0^2 z_0\partial^2 z_0 \quad (\partial w_0)^2 \\ x^2x'^2 - y^2y'^2 + z^2z'^2 - w^2w'^2 = s^2x_0^2 + z^2W^2 + z^2 - y^2 + z^2 - w^2, \\ x^2y'^2 - y^2x'^2 + z^2w'^2 - w^2z'^2 = -y^2x_0^2 - w^2z_0^2 - y^2 + x^2 - w^2 + z^2, \\ x^2z'^2 - y^2w'^2 + z^2x'^2 - w^2y'^2 = z^2x_0^2 + x^2z_0^2 + z^2 - w^2 + x^2 - y^2, \\ x^2w'^2 - y^2z'^2 + z^2y'^2 - w^2x'^2 = -w^2x_0^2 - y^2z_0^2 - w^2 + z^2 - y^2 + x^2. \end{array}$$

135. On substituting these values, the constant terms (or terms independent of u', v') disappear of themselves; and the equations, transposing the second and third of them, become

$$\begin{array}{rl} & \qquad\qquad\qquad\qquad x_0\partial^2 x_0 \quad (\partial y_0)^2 \quad z_0\partial^2 z_0 \quad (\partial w_0)^2 \\ (z_0^2 - x_0^2)\{x\partial^2 x - (\partial x)^2\} = & (-x_0^2x^2 + z_0^2z^2) + (x_0^2y^2 - z_0^2w^2) + (-x_0^2z^2 + z_0^2x^2) + (x_0^2w^2 - z_0^2y^2), \\ ,, \quad \{y\partial^2 y - (\partial y)^2\} = & -(x_0^2y^2 - z_0^2w^2) - (-x_0^2x^2 + z_0^2z^2) - (x_0^2w^2 - z_0^2y^2) - (-x_0^2z^2 + z_0^2x^2), \\ ,, \quad \{z\partial^2 z - (\partial z)^2\} = & (-x_0^2z^2 + z_0^2x^2) + (x_0^2w^2 - z_0^2y^2) + (-x_0^2x^2 + z_0^2z^2) + (x_0^2y^2 - z_0^2w^2), \\ ,, \quad \{w\partial^2 w - (\partial w)^2\} = & -(x_0^2w^2 - z_0^2y^2) - (-x_0^2z^2 + z_0^2x^2) - (x_0^2y^2 - z_0^2w^2) - (-x_0^2x^2 + z_0^2z^2). \end{array}$$

where it will be recollected that x, y, z, w mean $\vartheta_4, \vartheta_7, \vartheta_8, \vartheta_{11}(u)$; x_0 is $\vartheta_4(0)$, that is, c_4, and z_0 is $\vartheta_8(0)$, that is, c_8. But the formulae contain also

$$\partial^2 x_0 = (c_4''', c_4'''', c_4'' \mid u', v')^2, \quad \partial y_0 = (c_7', c_7'' \mid u', v'),$$

$$\partial^2 z_0 = (c_8''', c_8'''', c_8'' \mid u', v')^2, \quad \partial w_0 = (c_{11}', c_{11}'' \mid u', v').$$

The formulae may be written

$$\begin{array}{rl} & \qquad\qquad c_4\partial^2 c_4 \quad (\partial c_7)^2 \quad c_8\partial^2 c_8 \quad (\partial c_{11})^2 \\ (c_8^2 - c_4^2)\{4\partial^2 4 - (\partial 4)^2\} = & (-4\cdot 4 + 8\cdot 8) + (4\cdot 7 - 8\cdot 11) + (-4\cdot 8 + 8\cdot 4) + (4\cdot 11 - 8\cdot 7), \\ ,, \quad \{7\partial^2 7 - (\partial 7)^2\} = & -(4\cdot 7 - 8\cdot 11) - (-4\cdot 4 + 8\cdot 8) - (4\cdot 11 - 8\cdot 7) - (-4\cdot 8 + 8\cdot 4), \\ ,, \quad \{8\partial^2 8 - (\partial 8)^2\} = & (-4\cdot 8 + 8\cdot 4) + (4\cdot 11 - 8\cdot 7) + (-4\cdot 4 + 8\cdot 8) + (4\cdot 7 - 8\cdot 11), \\ ,, \quad \{11\partial^2 11 - (\partial 11)^2\} = & -(4\cdot 11 - 8\cdot 7) - (-4\cdot 8 + 8\cdot 4) - (4\cdot 7 - 8\cdot 11) - (-4\cdot 4 + 8\cdot 8). \end{array}$$

where $\partial^2 c_4$, $\partial^2 c_8$, ∂c_7, ∂c_{11} are written in place of $\partial^2 x_0$, $\partial^2 z_0$, ∂y_0, ∂w_0. There is of course a like system of equations for each of the Göpel tetrads.

136. Observe that, dividing the first equation by $\vartheta_4^2(u)$ or say by ϑ_4^2, the left-hand side is a mere constant multiple of $\partial^2 \log \vartheta_4$; and the right-hand side depends only on the quotient-functions $\vartheta_7 \div \vartheta_4$, $\vartheta_8 \div \vartheta_4$, $\vartheta_{11} \div \vartheta_4$; each side is a quadric function of u', v'. Equating the terms in u'^2, $u'v'$, v'^2 respectively, we have

$$\frac{d^2}{du^2}\log\vartheta_4,\quad \frac{d^2}{du\,dv}\log\vartheta_4,\quad \frac{d^2}{dv^2}\log\vartheta_4,$$

each of them expressed as a linear function of the squares of the quotient-functions $\vartheta_7 \div \vartheta_4$, $\vartheta_8 \div \vartheta_4$, $\vartheta_{11} \div \vartheta_4$. The formula is thus a second-derivative formula serving for the expression of a double theta-function by means of three quotient-functions.

Differential relations of the theta-functions.

137. In "The second set of 16," selecting the eight equations which contain Y_1 and W_1, these are

$u+u' \quad u-u' \quad u+u' \quad u-u'$	(Suffixes 1.)	
$\frac{1}{2}\{\vartheta_4 \cdot \vartheta_4 - \vartheta_{12} \cdot \vartheta_8\} \cdot \vartheta_8\vartheta_{12}$	$= Y' + W'$,	$Y \quad W$
$\frac{1}{2}\{\vartheta_{12}\vartheta_8 - \vartheta_8\vartheta_{12}\}$	$= Y' - W'$,	
$\frac{1}{2}\{\vartheta_6\vartheta_2 - \vartheta_2\vartheta_6\}$	$= W' + Y'$,	
$\frac{1}{2}\{\vartheta_{14}\vartheta_{10} - \vartheta_{10}\vartheta_{14}\}$	$= W' - Y'$,	
$\frac{1}{2}\{\vartheta_5\vartheta_1 + \vartheta_1\vartheta_5\}$	$= X' + Z'$,	$X \quad Z$
$\frac{1}{2}\{\vartheta_{13}\vartheta_9 + \vartheta_9\vartheta_{13}\}$	$= X' - Z'$,	
$\frac{1}{2}\{\vartheta_7\vartheta_3 + \vartheta_3\vartheta_7\}$	$= Z' + X'$,	
$\frac{1}{2}\{\vartheta_{15}\vartheta_{11} + \vartheta_{11}\vartheta_{15}\}$	$= Z' - X'$.	

Then, considering any line in the upper half and any two lines in the lower half, we can from the three equations eliminate Y_1 and W_1, thus obtaining an equation such as

$$\begin{vmatrix} \vartheta_4\vartheta_4 - \vartheta_{12}\vartheta_8 & Y_1' & W_1' \\ \vartheta_5\vartheta_1 + \vartheta_1\vartheta_5 & X_1' & Z_1' \\ \vartheta_7\vartheta_3 + \vartheta_3\vartheta_7 & X_1' & Z_1' \end{vmatrix} = 0,$$

viz. this is

$$-2X'Z'(\vartheta_4\vartheta_0 - \vartheta_0\vartheta_4) + (Z'^2 + X'^2)(\vartheta_5\vartheta_1 + \vartheta_1\vartheta_5) + (-Z'^2 + X'^2)(\vartheta_7\vartheta_3 + \vartheta_3\vartheta_7) = 0,$$

where the arguments of the theta-functions are as above, $u+u',\ u-u',\ u+u',\ u-u'$; and the suffixes of the X', Y', Z', W' are all $= 1$.

138. Suppose that in this equation u' becomes indefinitely small. If u' were $= 0$, the values of X', Y', Z', W' would be $\alpha, 0, \gamma, 0$: hence u' being indefinitely small, we take them to be $\alpha, \partial\beta, \gamma, \partial\delta$, where

$$\partial\beta = \left(u'\frac{d}{du} + v'\frac{d}{dv}\right)\beta, \qquad \partial\delta = \left(u'\frac{d}{du} + v'\frac{d}{dv}\right)\delta,$$

are, in fact, linear functions of u' and v'.

We have $\vartheta_4\vartheta_0 - \vartheta_0\vartheta_4$, standing for

$$\vartheta_4(u+u')\,\vartheta_0(u-u') - \vartheta_0(u+u')\,\vartheta_4(u-u'),$$

and here

$$\vartheta_4(u \pm u') = \vartheta_4 \pm \partial\vartheta_4, \quad \vartheta_0(u \pm u') = \vartheta_0 \pm \partial\vartheta_0;$$

the function in question is thus

$$(\vartheta_4 + \partial\vartheta_4)(\vartheta_0 - \partial\vartheta_0) - (\vartheta_4 - \partial\vartheta_4)(\vartheta_0 + \partial\vartheta_0) = 2(\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0),$$

where the arguments are u, v, and the ∂ denotes $u'\dfrac{d}{du} + v'\dfrac{d}{dv}$.

Also $\vartheta_5\vartheta_1 + \vartheta_1\vartheta_5$, that is, $\vartheta_5(u+u')\vartheta_1(u-u') + \vartheta_1(u+u')\vartheta_5(u-u')$, becomes simply $= 2\vartheta_5\vartheta_1$, and similarly $\vartheta_7\vartheta_3 + \vartheta_3\vartheta_7$ becomes $= 2\vartheta_{13}\vartheta_9$ [as in scan]; and the equation thus is

$$-2\alpha_1\gamma_1(\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0) + (\alpha_1\partial\delta_1 + \gamma_1\partial\beta_1)\vartheta_5\vartheta_1 + (-\alpha_1\partial\beta_1 + \gamma_1\partial\delta_1)\vartheta_3\vartheta_7 = 0,$$

where the proper suffix 1 is restored to the $\alpha,\ \partial\beta,\ \gamma$, and $\partial\delta$.

139. The equation shows that the differential combination $\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0$ is a linear function of $\vartheta_5\vartheta_1$ and $\vartheta_3\vartheta_7$, the coefficients of these products being of course linear functions of u' and v'. Writing the equation

$$\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0 = A\vartheta_5\vartheta_1 + B\vartheta_3\vartheta_7,$$

we can if we please determine the coefficients in terms of the constants c', c'', c''', c'''', c''; viz. taking u, v indefinitely small, we have

$$\vartheta_0 = c_0, \quad \partial\vartheta_0 = u'(c_0'''u + c_0''''v) + v'(c_0''''u + c_0''v),$$

$$\vartheta_4 = c_4, \quad \partial\vartheta_4 = u'(c_4'''u + c_4''''v) + v'(c_4''''u + c_4''v),$$

$$\vartheta_1 = c_1, \quad \vartheta_5 = c_5'u + c_5''v,$$

or substituting, and equating the coefficients of u and v respectively, we have

$$c_0(c_4'''u' + c_4''''v') - c_4(c_0'''u' + c_0''''v') = Ac_5c_1' + Bc_3c_7',$$

$$c_0(c_4''''u' + c_4''v') - c_4(c_0''''u' + c_0''v') = Ac_5c_1'' + Bc_3c_7'',$$

which equations give the values of A, B.

140. Disregarding the values of the coefficients, and attending only to the form of the equation

$$\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0 = A\vartheta_5\vartheta_1 + B\vartheta_3\vartheta_7,$$

this is one of a system of 120 equations; viz. referring to the foregoing table of the 120 pairs, it in fact appears that taking any pair such as $\vartheta_0\vartheta_4$ out of the upper compartment or the lower compartment of any column of the table, the corresponding differential combination $\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0$ is a linear function of any two of the four pairs in the other compartment of the same column.

Differential relation of u, v and x, y.

141. We have, as before, in the two notations, the pairs

$A \cdot B$	$11 \cdot 7$	
$C \cdot DE$	5	9
$D \cdot CE$	13	1
$E \cdot CD$	14	2
$F \cdot AB$	10	6

From the expressions given above for the four pairs below the line, it is clear that any linear function of these four pairs may be represented by

$$(a-b)\tfrac{1}{\theta}\{(\lambda+\mu y)\sqrt{cdefa_2b_2} + (\lambda+\mu x)\sqrt{c_2d_2e_2f_2ab}\},$$

where λ, μ are constant coefficients: the factor $(a-b)$ has been introduced for convenience, as will appear.

We have consequently a relation

$$\sqrt{aa_2}\,\partial\sqrt{bb_2} - \sqrt{bb_2}\,\partial\sqrt{aa_2} = \frac{a-b}{\theta}\{(\lambda+\mu y)\sqrt{cdefa_2b_2} + (\lambda+\mu x)\sqrt{c_2d_2e_2f_2ab}\},$$

where, as before, ∂ is used to denote $u'\dfrac{d}{du} + v'\dfrac{d}{dv}$, u' and v' being arbitrary multipliers; considering u, v as functions of x, y, we have

$$\frac{d}{du} = \frac{dx}{du}\frac{d}{dx} + \frac{dy}{du}\frac{d}{dy}, \qquad \frac{d}{dv} = \frac{dx}{dv}\frac{d}{dx} + \frac{dy}{dv}\frac{d}{dy},$$

and thence $\partial = P\dfrac{d}{dx} + Q\dfrac{d}{dy}$ if for shortness P and Q are written to denote $u'\dfrac{dx}{du} + v'\dfrac{dx}{dv}$ and $u'\dfrac{dy}{du} + v'\dfrac{dy}{dv}$ respectively.

142. The left-hand side then is

$$= P\left(\sqrt{aa_2}\,\frac{d}{dx}\sqrt{bb_2} - \sqrt{bb_2}\,\frac{d}{dx}\sqrt{aa_2}\right) + Q\left(\sqrt{aa_2}\,\frac{d}{dy}\sqrt{bb_2} - \sqrt{bb_2}\,\frac{d}{dy}\sqrt{aa_2}\right);$$

the coefficients of P and Q are at once found to be

$$\tfrac{1}{2}\frac{(a-b)\sqrt{a_2b_2}}{\sqrt{ab}}, \qquad \tfrac{1}{2}\frac{(a_2-b_2)\sqrt{ab}}{\sqrt{a_2b_2}},$$

respectively, or observing that $a-b_2=a_2-b_2=a-b$, the equation becomes

$$P\frac{\sqrt{a_2b_2}}{\sqrt{ab}}+Q\frac{\sqrt{ab}}{\sqrt{a_2b_2}}=\frac{2}{\theta}\big\{(\lambda+\mu y)\sqrt{cdefa_2b_2}+(\lambda+\mu x)\sqrt{c_2d_2e_2f_2ab}\big\},$$

or multiplying by $\sqrt{aba_2b_2}$ and writing for shortness $abcdef=X$, $a_2b_2c_2d_2e_2f_2=Y$, this becomes

$$a_2b_2\big\{P+\tfrac{2}{\theta}(\lambda+\mu y)\sqrt{X}\big\}+ab\big\{Q+\tfrac{2}{\theta}(\lambda+\mu x)\sqrt{Y}\big\}=0.$$

143. There are, it is clear, the like equations

$$b_2c_2\big\{P+\tfrac{2}{\theta}(\lambda'+\mu' y)\sqrt{X}\big\}+bc\big\{Q+\tfrac{2}{\theta}(\lambda'+\mu' x)\sqrt{Y}\big\}=0,$$

$$c_2a_2\big\{P+\tfrac{2}{\theta}(\lambda''+\mu'' y)\sqrt{X}\big\}+ca\big\{Q+\tfrac{2}{\theta}(\lambda''+\mu'' x)\sqrt{Y}\big\}=0,$$

and it is to be shown that $\lambda=\lambda'=\lambda''$ and $\mu=\mu'=\mu''$. For this purpose, recurring to the forms

$$\sqrt{aa_2}\,\partial\sqrt{bb_2}-\sqrt{bb_2}\,\partial\sqrt{aa_2}=\tfrac{a-b}{\theta}\big\{(\lambda+\mu y)\sqrt{cdefa_2b_2}+(\lambda+\mu x)\sqrt{c_2d_2e_2f_2ab}\big\},$$

$$\sqrt{bb_2}\,\partial\sqrt{cc_2}-\sqrt{cc_2}\,\partial\sqrt{bb_2}=\tfrac{b-c}{\theta}\big\{(\lambda'+\mu' y)\sqrt{adefb_2c_2}+(\lambda'+\mu' x)\sqrt{a_2d_2e_2f_2bc}\big\},$$

$$\sqrt{cc_2}\,\partial\sqrt{aa_2}-\sqrt{aa_2}\,\partial\sqrt{cc_2}=\tfrac{c-a}{\theta}\big\{(\lambda''+\mu'' y)\sqrt{bdefc_2a_2}+(\lambda''+\mu'' x)\sqrt{b_2d_2e_2f_2ca}\big\},$$

multiply the first equation by $\sqrt{cc_2}$, the second by $\sqrt{aa_2}$, and the third by $\sqrt{bb_2}$, and add: the left-hand side vanishes, and therefore the right-hand side must also vanish identically.

144. But on the right-hand side we have the term $\frac{1}{\theta}\sqrt{defa_2b_2c_2}$ multiplied by

$$(a-b)c(\lambda+\mu y)+(b-c)a(\lambda'+\mu' y)+(c-a)b(\lambda''+\mu'' y),$$

and the term $-\frac{1}{\theta}\sqrt{d_2e_2f_2abc}$ multiplied by

$$(a-b)c_2(\lambda+\mu x)+(b-c)a_2(\lambda'+\mu' x)+(c-a)b_2(\lambda''+\mu'' x),$$

and it is clear that the whole can vanish only if these two coefficients separately vanish. This will be the case if we have for $\lambda, \lambda', \lambda''$ the equations

$$(a-b)\lambda+(b-c)\lambda'+(c-a)\lambda''=0,$$

$$c \quad ,,+a \quad ,,+b \quad ,,=0,$$

and the like equations for μ, μ', μ''. The equations written down give

$$(a-b)\lambda : (b-c)\lambda' : (c-a)\lambda'' = a-b : b-c : c-a,$$

that is, $\lambda = \lambda' = \lambda''$: and similarly $\mu = \mu' = \mu''$.

145. But this being so, the three equations in P, Q give

$$P + \tfrac{2}{\theta}(\lambda + \mu y)\sqrt{X} = 0, \quad Q + \tfrac{2}{\theta}(\lambda + \mu x)\sqrt{Y} = 0,$$

that is,

$$-\tfrac{1}{2}\theta\,\frac{dx}{\sqrt{X}} = (\lambda + \mu y)\,du + (\lambda + \mu y)\,dv \quad \text{[as in scan]},$$

$$-\tfrac{1}{2}\theta\,\frac{dy}{\sqrt{Y}} = (\lambda + \mu x)\,du + (\lambda + \mu x)\,dv.$$

In these equations u' and v' are arbitrary; hence λ and μ must be linear functions of u' and v'; say their values are $= \varpi u' + \rho v',\ \sigma u' + \tau v'$ respectively. We have therefore

$$-\tfrac{1}{2}\theta\,\frac{dx}{\sqrt{X}} = (\varpi + \sigma y)\,du + (\rho + \tau y)\,dv,$$

$$-\tfrac{1}{2}\theta\,\frac{dy}{\sqrt{Y}} = (\varpi + \sigma x)\,du + (\rho + \tau x)\,dv,$$

whence also

$$\varpi\,du + \rho\,dv = -\tfrac{\theta}{2}\left(\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}}\right),$$

$$\sigma\,du + \tau\,dv = -\tfrac{\theta}{2}\left(\frac{x\,dx}{\sqrt{X}} + \frac{y\,dy}{\sqrt{Y}}\right),$$

which are the required relations, depending on the square roots of the sextic functions $X = abcdef$, and $Y = a_2b_2c_2d_2e_2f_2$, of x and y respectively; but containing the constants $\varpi, \rho, \sigma, \tau$, the values of which are not as yet ascertained.

146. I commence the integration of these equations on the assumption that the values $u=0,\ v=0$ correspond to indefinitely large values of x and y. We have

$$\sqrt{X} = x^3\left(1 - \tfrac{S}{x} + \ldots\right), \qquad \sqrt{Y} = y^3\left(1 - \tfrac{S}{y} + \ldots\right),$$

where $S = a+b+c+d+e+f$; and thence the equations are

$$\varpi\,du + \rho\,dv = -\tfrac{\theta}{2}\left[\frac{dx}{x^3}\left(1 + \tfrac{S}{x} + \ldots\right) + \frac{dy}{y^3}\left(1 + \tfrac{S}{y} + \ldots\right)\right],$$

Hence integrating, we have

$$\varpi u + \rho v = \tfrac{\theta}{4}\left(\frac{1}{x^2} + \frac{1}{y^2}\right) + \ldots,$$

and thence

$$\sigma u + \tau v = \tfrac{\theta}{2}\left(\frac{1}{x} + \frac{1}{y}\right) + \tfrac{1}{2}S\left(\frac{1}{x^2} + \frac{1}{y^2}\right) + \ldots,$$

where the omitted terms depend on $\frac{1}{x^3}, \frac{1}{y^3}$, &c.

Hence, neglecting these terms, we have

$$\frac{\sigma u + \tau v}{\varpi u + \rho v + \frac{1}{2}S(\sigma u + \tau v)} = -\left(\frac{1}{x} + \frac{1}{y}\right),$$

an equation connecting the indefinitely small values of u, v, with the indefinitely large values of x, y.

147. From the equations $A = \frac{1}{2}k_3\varpi\sqrt{a}$, $B = \frac{1}{2}k_3\sqrt{b}$, taking (u, v) indefinitely small and therefore (x, y) indefinitely large, we deduce

$$\frac{c'_{11}u + c''_{11}v}{c'_7 u + c''_7 v} = \frac{k_a}{k_b} \cdot \frac{1 - \frac{1}{2}a\left(\frac{1}{x} + \frac{1}{y}\right)}{1 - \frac{1}{2}b\left(\frac{1}{x} + \frac{1}{y}\right)},$$

hence substituting for $\frac{1}{x} + \frac{1}{y}$ the foregoing value, and introducing an indeterminate multiplier M, we obtain

$$c'_{11}u + c''_{11}v = Mk_a\left[\varpi u + \rho v + \tfrac{1}{2}S(\sigma u + \tau v) + \tfrac{1}{2}a(\sigma u + \tau v)\right],$$

which breaks up into the two equations

[two component equations as in scan]

Similarly

$$\begin{aligned}
c'_7 &= Mk_b\{\ \ldots\ b\ \ldots\ \}, & c''_7 &= Mk_b\{\ \ldots\ b\ \ldots\ \},\\
c'_{10} &= Mk_c\{\ \ldots\ c\ \ldots\ \}, & c''_{10} &= Mk_c\{\ \ldots\ c\ \ldots\ \},\\
c'_{13} &= Mk_d\{\ \ldots\ d\ \ldots\ \}, & c''_{13} &= Mk_d\{\ \ldots\ d\ \ldots\ \},\\
c'_5 &= Mk_e\{\ \ldots\ e\ \ldots\ \}, & c''_5 &= Mk_e\{\ \ldots\ e\ \ldots\ \},\\
c'_{14} &= Mk_f\{\ \ldots\ f\ \ldots\ \}, & c''_{14} &= Mk_f\{\ \ldots\ f\ \ldots\ \},
\end{aligned}$$

which twelve equations determine the coefficients $\varpi, \sigma, \rho, \tau$ in terms of the c', c'' of the odd functions 5, 7, 10, 11, 13, 14; and moreover give rise to relations connecting these c', c'' with each other and with the constants a, b, c, d, e, f.

148. It is observed that if, as before,

$$\partial = u'\frac{d}{du} + v'\frac{d}{dv} = P\frac{d}{dx} + Q\frac{d}{dy},$$

then, substituting for P and Q their values, we have

$$\begin{aligned}\partial &= -\tfrac{\theta}{2}(\varpi u' + \rho v')\left(\sqrt{X}\frac{d}{dx} + \sqrt{Y}\frac{d}{dy}\right) - \tfrac{\theta}{2}(\sigma u' + \tau v')\left(x\sqrt{X}\frac{d}{dx} + y\sqrt{Y}\frac{d}{dy}\right)\\ &= (\varpi u' + \rho v')\partial_1 + (\sigma u' + \tau v')\partial_2,\end{aligned}$$

if for shortness

$$\partial_1 = -\tfrac{\theta}{2}\left(\sqrt{X}\frac{d}{dx} + \sqrt{Y}\frac{d}{dy}\right), \qquad \partial_2 = -\tfrac{\theta}{2}\left(x\sqrt{X}\frac{d}{dx} + y\sqrt{Y}\frac{d}{dy}\right);$$

then operating with ∂ on the equations $A = \varpi k_a\sqrt{a}$, &c., we have for instance

$$A\partial B - B\partial A = \varpi^2 k_a k_b\big\{(\varpi u' + \rho v')(\sqrt{a}\,\partial_1\sqrt{b} - \sqrt{b}\,\partial_1\sqrt{a}) + (\sigma u' + \tau v')(\sqrt{a}\,\partial_2\sqrt{b} - \sqrt{b}\,\partial_2\sqrt{a})\big\},$$

which is one of a system of 120 equations, the A, B being in fact any two of the 16 functions.

These are in fact nothing else than the foregoing system of 120 equations giving the values of the differential combinations $\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0$, &c., each as a sum of products of pairs of functions, only on the right-hand sides we have expressions such as $\sqrt{a}\,\partial_1\sqrt{b} - \sqrt{b}\,\partial_1\sqrt{a}$, &c., which present themselves as perfectly determinate functions of x, y: so that, regarding $\varpi u' + \rho v'$, $\sigma u' + \tau v'$ as given linear functions of the arbitrary quantities u', v', there is no longer anything indeterminate in the form of the equations.

705.

PROBLEMS AND SOLUTIONS.

[From the Mathematical Questions with their Solutions from the Educational Times, vols. XIV. to LXI. (1871—1894).]

[Vol. XIV., July to December, 1870, pp. 17—19.]

3002. (Proposed by Matthew Collins, B.A.)—If every two of five circles A, B, C, D, E touch each other, except D and E, prove that the common tangent of D and E is just twice as long as it would be if D and E touched each other.

Solution by Professor Cayley.

Consider the ellipse $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, foci R, S; the coordinates of a point U on the ellipse may be taken to be $(a\cos u, b\sin u)$, and then the distances of this point from the foci will be

$$r = a(1 - e\cos u), \qquad s = a(1 + e\cos u).$$

Taking k arbitrarily, with centre R describe a circle radius $a - k$, with centre S a circle radius $a + k$, and with centre U a circle radius $k - ae\cos u$; say these are the circles R, S, U respectively; the circle U will touch each of the circles R, S (viz. assuming $ae < k < a$, so that the foregoing radii are all positive, it will touch the circle R externally and the circle S internally).

Considering next a point V, coordinates $(a\cos v, b\sin v)$, and the circle described about this point with the radius $k - ae\cos v$, say the circle V; this will touch in like manner the circles R, S respectively. And the circles U, V may be made to touch each other externally; viz. this will be the case if squared sum of radii = squared...